Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

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Abstract

In nonlinear model predictive control (NMPC), sequential quadratic programming (SQP) with the exact Hessian of the Lagrangian produces subproblems that are quadratic programs (QPs) with a potentially indefinite full Hessian. Structure-exploiting QP solvers embedded in the SQP solver often require a positive definite full Hessian, so the full Hessian is typically regularized at the SQP level before it is passed to the QP solver, which in general changes the solution of each QP and also the feedback in SQP with real-time iteration. In contrast to SQP-level regularizations, regularizing at the QP level in interior point method (IPM) can introduce two advantages. First, this lets the QP solver exploit the curvature from inequality constraints without extra estimation, and, second, only return the solution of the original QP.

This thesis investigates primal regularizations inside interior-point solvers for indefinite QPs with optimal control problem (OCP) structure, with a focus on the proposed Riccati-based regularization. This thesis shows that the minimum perturbation strategy can make the Riccati-based regularization fail, and, based on these observations, an improved strategy is derived to avoid such failure cases. It is proven that the Riccati-based regularization becomes inactive near the solution, as it regularizes only at the stages where the Cholesky factorization fails during Riccati recursion.

For evaluation, this thesis builds a benchmark of 170 indefinite OCP QPs from SQP iterations when solving NMPC problems. With the regularizations implemented in a prototype solver that reproduces the iterations of HPIPM on convex QPs, this thesis compares the Riccati-based regularization with full Hessian regularization and inertia correction in terms of robustness, efficiency and estimated factorization cost.

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Acronyms

Acronyms	Description
FTB	Fraction-to-the-boundary
GMW	Gill–Murray–Wright
IPM	Interior point method
KKT	Karush–Kuhn–Tucker
LHS	Left-hand side
LICQ	Linear independence constraint qualification
NLP	Nonlinear programming problem
NMPC	Nonlinear model predictive control
OCP	Optimal control problem
QP	Quadratic program
RHS	Right-hand side
SONC	Second-order necessary condition
SOSC	Second-order sufficient condition
SQP	Sequential quadratic programming

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Chapter 1

Introduction

1.1 Motivation

In nonlinear model predictive control (NMPC), a nonlinear optimal control problem (OCP) is solved in each sampling period to compute the optimal control input for the current state. To solve the continuous-time OCPs, direct multiple shooting [1] is a widely used method, which transcribes the OCP into a nonlinear programming problem (NLP) by discretizing the time horizon into multiple shooting intervals [2]. Sequential quadratic programming (SQP) [3] provides a suitable approach for solving the NLP, which linearizes to obtain a quadratic program (QP) subproblem and solves it to obtain the search direction at each iterate. In the real-time iteration [4], one SQP iteration is performed per sampling time, with the linearization done before the new state is known, so only the QP solve lies between measurement and feedback. The QP solve therefore requires efficiency, for which structure-exploiting solvers are usually employed.

When the exact Hessian of the Lagrangian is used in QP subproblems [5], they might have an indefinite full Hessian. The QP solvers embedded in SQP for exploiting OCP structure, such as qpDUNES [6], FORCES [7] and HPIPM [8], however, may fail to solve these QPs, for they require convex QPs with a positive definite full Hessian. One method to use these solvers in this situation is to regularize the full Hessian at the SQP level before it is passed to the solver, while preserving the OCP structure. Two common regularization methods mentioned in [9], i.e. project and mirror, are implemented in acados [10] and used to yield a

positive definite full Hessian for the QP solver. Both methods regularize based on the eigenvalues of the full Hessian and use a small positive value as a threshold. Project raises every eigenvalue to at least the threshold, while mirror replaces the negative eigenvalues with their absolute values, using the threshold as a lower bound. This, however, in general modifies the solution of the subproblems and also the feedback itself in the real-time iteration.

Convexification [9], as a more advanced regularization method, applies a backward sweep similar to a Riccati recursion for regularization. The sweep transfers cost between neighboring stages through the dynamics and adds curvature from the estimated active inequality constraints with an estimated weight to each stage. The cost transfer keeps only the needed curvature for each stage to be positive definite while leaving the reduced Hessian, i.e. the Hessian on the null space of the dynamics, unchanged. The added curvature makes the reduced Hessian positive definite if the estimation of the active constraints and weight is correct and the fully reduced Hessian, i.e. the Hessian on the null space of the dynamics and the active constraints, is positive definite. This results in a positive definite full Hessian with the fully reduced Hessian unchanged, and the same primal solution of the original QP is thus kept. But if the fully reduced Hessian is indefinite or the estimation is incorrect, the backward sweep could still reach a stage with indefinite blocks, and convexification falls back to projecting the stage block, which modifies the QP solution. In acados, only the cost transfer in convexification is implemented, so it could fail to regularize with inequality constraints in practice.

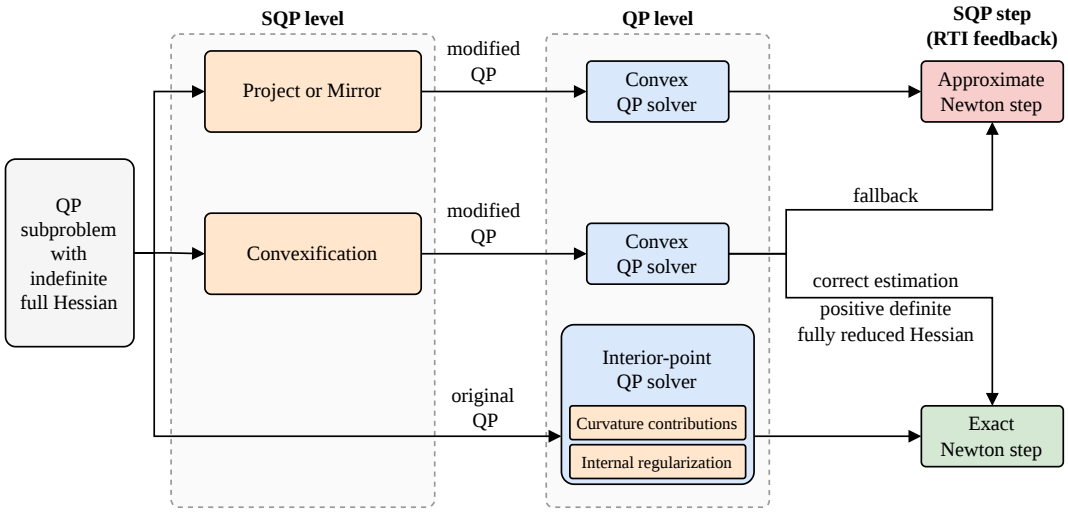


Figure 1.1: Regularization overview of an indefinite QP subproblem

In an interior-point QP solver, however, the interior point method (IPM) relaxes the complementarity from each inequality constraint by a barrier parameter. This adds weighted curvature from inequality constraints to the Hessian inherently throughout the iterations. The weight on each constraint, i.e. the ratio of its multiplier to its slack, changes with the current iterate, so neither an estimate of the active set nor a penalty weight is needed. The reduced Hessian with curvature from inequality constraints is positive definite, as long as the fully reduced Hessian is positive definite.

Example 1 (A QP with an indefinite reduced Hessian)**.** Consider an indefinite QP

$$\begin{aligned} \underset{x_0,\, u_0,\, x_1,\, u_1,\, x_2}{\text{minimize}} \quad & \tfrac{1}{2}(u_0^2 - 3x_1^2 + u_1^2 + \tfrac{1}{2}x_2^2) \\ \text{subject to} \quad & x_0 = 1, \\ & x_1 = x_0 + u_0, \\ & x_2 = x_1 + u_1, \\ & -1 \le u_0 \le 1. \end{aligned} \tag{1.1}$$

By eliminating x_0, x_1 and x_2 using the equality constraints, we have a QP with an indefinite reduced Hessian as:

$$\begin{aligned} \underset{u_0,\, u_1}{\text{minimize}} \quad & \frac{1}{2}\begin{bmatrix} u_0 \\ u_1 \end{bmatrix}^\top \begin{bmatrix} -\frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \end{bmatrix} + \begin{bmatrix} -\frac{5}{2} \\ \frac{1}{2} \end{bmatrix}^\top \begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \\ \text{subject to} \quad & -1 \le u_0 \le 1. \end{aligned} \tag{1.2}$$

Denote the objective of (1.2) by $f(u_0, u_1)$. For a fixed u_0, the minimizer of the objective over u_1 follows from

$$\frac{\partial f}{\partial u_1} = \tfrac{1}{2}u_0 + \tfrac{3}{2}u_1 + \tfrac{1}{2} = 0 \quad \Longrightarrow \quad u_1^\star(u_0) = -\tfrac{1}{3}(u_0 + 1).$$

Substituting $u_1^\star(u_0)$ back into f leaves a function of u_0 alone,

$$f\Big(u_0, u_1^\star\Big) = -\tfrac{5}{6}u_0^2 - \tfrac{8}{3}u_0 - \tfrac{1}{12}.$$

Since $f\Big(u_0, u_1^\star\Big)$ is concave, its minimum over $-1 \le u_0 \le 1$ lies on a bound, and comparing $f\Big(-1, 0\Big) = \frac{7}{4}$ with $f\Big(1, -\frac{2}{3}\Big) = -\frac{43}{12}$ gives the exact solution $(1, -\frac{2}{3})$.

The contour lines of the reduced objective are shown in Figure 1.2a with a saddle point in reduced space and the exact solution lying at $(1, -\frac{2}{3})$ on the constraint boundary.

From an SQP perspective, this QP can occur either as a subproblem within the SQP iterations or as an NLP that is then solved using SQP.

In the former case, the SQP-level regularization, here mirror, modifies the QP before it is passed to the QP solver. With decision variables as $(x_0, u_0, x_1, u_1, x_2)$, the full Hessian of (1.1) is $\operatorname{diag}\left(0, 1, -3, 1, \frac{1}{2}\right)$. Mirror replaces the eigenvalue -3 by 3 and lifts 0 to a threshold 10^{-4}, resulting in the regularized full Hessian as $\operatorname{diag}\left(10^{-4}, 1, 3, 1, \frac{1}{2}\right)$. Eliminating the states as before gives the reduced subproblem

$$\begin{aligned} \underset{u_0,\, u_1}{\text{minimize}} \quad & \frac{1}{2}\begin{bmatrix} u_0 \\ u_1 \end{bmatrix}^\top \begin{bmatrix} \frac{9}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \end{bmatrix} + \begin{bmatrix} \frac{7}{2} \\ \frac{1}{2} \end{bmatrix}^\top \begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \\ \text{subject to} \quad & -1 \leq u_0 \leq 1, \end{aligned}$$

whose reduced Hessian becomes positive definite, and the corresponding solution is $(-\frac{10}{13}, -\frac{1}{13})$ shown in Figure 1.2b. It is approximate for the QP subproblem and also an approximate Newton step for SQP.

For completeness, Figure 1.2c shows the latter case, where SQP solves (1.1) as an NLP. SQP regularizes and linearizes at each iterate, for a QP, the regularized Hessian stays constant, only the gradient changes. From the cold start, SQP first takes the approximate step from the first subproblem and then takes several iterations to reach the exact solution, which is the fixed point satisfying the optimality conditions of (1.1).

From a QP perspective, the curvature contributions from inequality constraints can be exploited in the IPM. At the cold start, both bounds in $-1 \leq u_0 \leq 1$ have the slack as 1. Choosing the initial barrier parameter as 10, the corresponding initialized multiplier is 10, so the summed weight of two constraints is 20. Since both bounds act on u_0 only, it adds to the u_0 entry of the reduced Hessian,

$$\begin{bmatrix} -\frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} + \begin{bmatrix} 20 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{37}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix},$$

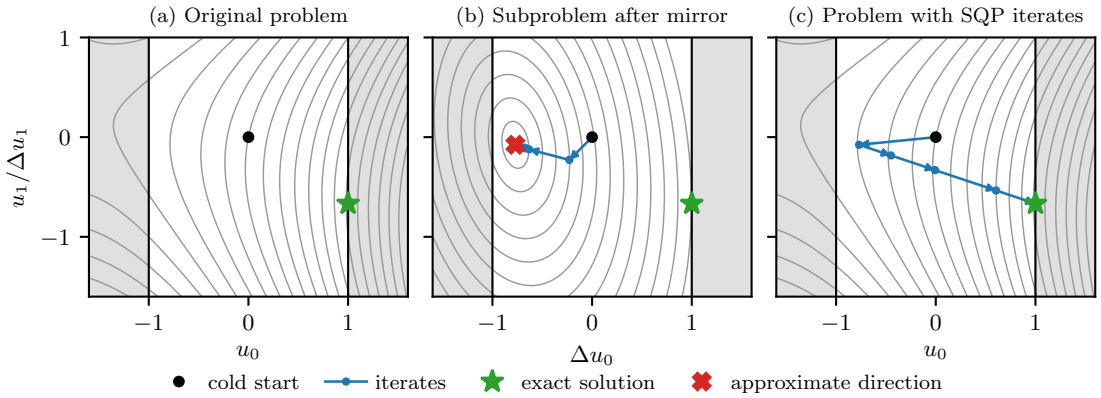


Figure 1.2: Contour lines of a QP with an indefinite reduced Hessian. (a) shows the original objective, (b) shows the subproblem objective of the QP after mirror, and (c) shows the SQP iterates when the QP is solved as an NLP for completeness.

which is positive definite. The local second-order model in Figure 1.3b is therefore convex without any extra work, the IPM can thus solve the indefinite QP in several QP iterations, as shown in Figure 1.3c.

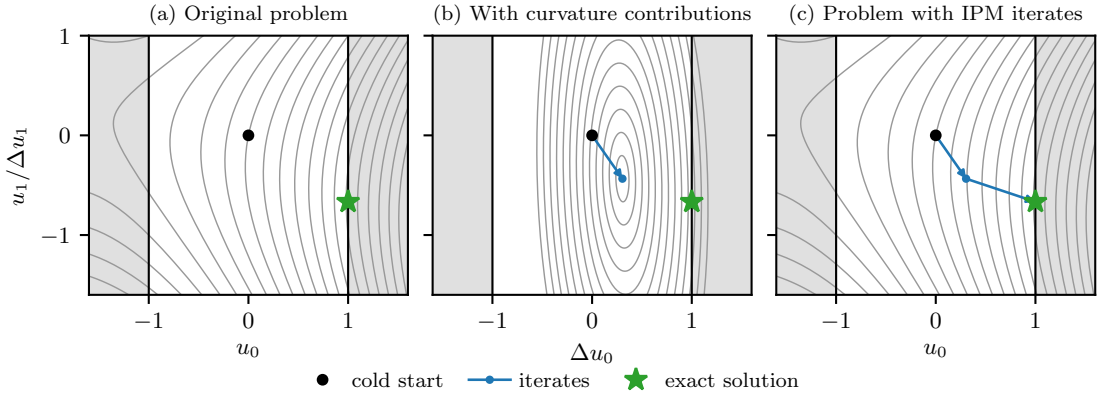


Figure 1.3: Contour lines of a QP with an indefinite reduced Hessian. (a) shows the original objective, (b) shows the local second-order model with curvature contributions from inequality constraints, and (c) shows the IPM iterates on the QP.

In other cases, such as in the early SQP iterations, where the curvature contribution may not be enough to make the reduced Hessian positive definite, a regularization to the Hessian is still needed. But if the regularization is made inside the solver, the solution of the original QP remains, and thus an exact Newton step in SQP is preserved.

A structure-exploiting interior-point QP solver that handles indefiniteness at the QP level is therefore interesting as an alternative to regularizing at the SQP level. This consideration leads to the question of how the regularization should be performed inside the solver, and this thesis addresses this question for indefinite QPs with OCP structure.

1.2 Contributions

The main contributions of this thesis are the following:

1. **Riccati-based regularization**: This thesis investigates regularizations inside interior-point QP solvers. The main focus is the proposed Riccati-based regularization, which exploits the OCP structure of the problem, and regularizes the Hessian only when Cholesky factorization fails. This thesis further analyzes why the minimum perturbation in this regularization fails, improves the regularization strategy, and proves the regularization to be inactive near the solution (Chapter 3).
2. **A benchmark of 170 indefinite OCP QPs**: This thesis provides a pipeline for collecting QPs from SQP iterations performed within acados, and collects 170 QPs from 8 nonlinear OCPs, together with an analysis of the indefiniteness regarding the benchmark (Chapter 4).
3. **Numerical comparison and ablation**: This thesis implements a prototype solver that shows identical iteration behavior to HPIPM on convex QPs, and performs experiments with this solver on the benchmark to investigate the performance of the methods. In addition, this thesis presents an analysis of solving failures and includes ablation studies on the solver settings to show which components the results depend on (Chapter 5).

1.3 Outline

The remainder of this thesis is organized as follows. In Chapter 2, we start with theoretical background regarding primal–dual IPMs for OCP-structured QPs, where the Riccati recursion is used to solve the QPs. We introduce the barrier

update and step size control strategies adopted in this work. In Chapter 3, we propose the Riccati-based primal regularization, derive the improved stage-wise regularization strategy from the failure of minimum perturbation, and prove that the regularization is asymptotically inactive. Chapter 4 presents the benchmark of indefinite QPs constructed from the SQP iterations of eight OCPs, and analyzes how the indefiniteness is distributed over the stages and where it originates. Chapter 5 first describes the prototype solver, the experiment setup, and the numerical results. The experiments compare the Riccati-based regularizations with full Hessian regularization and inertia correction on the benchmark, and discuss the instances on which the methods fail, with ablations for investigating which components the results depend on. Finally, Chapter 6 summarizes the results and outlines directions for future work.

Chapter 2

Interior-point method for quadratic programs in optimal control

In this chapter, the theoretical background regarding the interior-point QP solvers is introduced. The primal–dual IPM is introduced first in Section 2.1 as the core algorithm for solving QPs in this thesis. Then, the QP with OCP structure is introduced in Section 2.2, where the Riccati recursion is applied for solving the QP with linear complexity. Finally, the adaptive barrier update and step size control strategies are discussed in Section 2.3 and Section 2.4 with consideration of the nonconvex case. Algorithm 1 outlines how these parts collaborate.

Algorithm 1 Primal–dual IPM for QPs with OCP structure

Input: QP with OCP structure, initial iterate, iteration budget $i_{\max}$
for $i = 0$ **to** $i_{\max} - 1$ **do**
    compute the residuals ▷ Section 2.1
    if all residuals are below their tolerances **then**
        return current iterate
    end if
    solve with Riccati recursion ▷ Section 2.2
    update barrier and solve again ▷ Section 2.3
    compute step sizes and update variables ▷ Section 2.4
end for

2.1 Primal–dual interior-point method

Consider a QP with equality and inequality constraints, of which the formulation can be written as

$$\begin{aligned} \underset{y}{\text{minimize}} \quad & \frac{1}{2} y^\top H y + g^\top y \\ \text{subject to} \quad & Fy = b, \\ & Gy \leq d, \end{aligned} \tag{2.1}$$

where y are the decision variables, H represents the full Hessian, g is the linear term, and F and G stand for the linear constraint matrices for equality and inequality constraints. Since the constraints are affine w.r.t. y, the definiteness of the full Hessian matrix H determines whether the QP in (2.1) is convex or nonconvex.

By introducing slack variables t, we have the equivalent formulation

$$\begin{aligned} \underset{y,\, t}{\text{minimize}} \quad & \frac{1}{2} y^\top H y + g^\top y \\ \text{subject to} \quad & Fy = b, \\ & Gy + t = d, \\ & t \geq 0. \end{aligned} \tag{2.2}$$

The Lagrangian function of the QP in (2.2) is

$$\mathcal{L}(y, t, \pi, \lambda) = \frac{1}{2} y^\top H y + g^\top y + \pi^\top (Fy - b) + \lambda^\top (Gy + t - d), \tag{2.3}$$

where π and $\lambda \geq 0$ are the Lagrange multipliers associated with the equality and inequality constraints, respectively. Then the Karush–Kuhn–Tucker (KKT) conditions of the QP in (2.2) are given by

$$Hy + g + F^\top \pi + G^\top \lambda = 0, \tag{2.4a}$$
$$Fy - b = 0, \tag{2.4b}$$
$$Gy + t - d = 0, \tag{2.4c}$$
$$\lambda_j \, t_j = 0, \quad j = 1, \ldots, n_c, \tag{2.4d}$$
$$\lambda \geq 0, \quad t \geq 0, \tag{2.4e}$$

where n_c is the number of inequality constraints. In the primal–dual IPM, the complementarity condition (2.4d) is reformulated as a smooth equation using the barrier parameter $\tau > 0$:

$$\lambda_j \, t_j = \tau, \quad j = 1, \ldots, n_c, \tag{2.5}$$

and the parameter τ is shrunk towards zero with a certain strategy during the iterations to obtain the solution of the original problem, and step size control is applied to ensure the satisfaction of the inequalities in (2.4e). The corresponding update strategy and step size control are discussed in the following sections.

In order to solve the nonlinear system in (2.4), Newton's method linearizes the system at each iteration i to obtain the step $(\Delta y, \Delta\pi, \Delta\lambda, \Delta t)$ from the linear system, which takes the form of

$$\begin{bmatrix} H & F^\top & G^\top & 0 \\ F & 0 & 0 & 0 \\ G & 0 & 0 & I \\ 0 & 0 & T_i & \Lambda_i \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta\pi \\ \Delta\lambda \\ \Delta t \end{bmatrix} = - \begin{bmatrix} r_i^{\text{stat}} \\ r_i^{\text{eq}} \\ r_i^{\text{ineq}} \\ r_i^{\text{comp}} \end{bmatrix}, \tag{2.6}$$

where the diagonal matrices $T_i = \operatorname{diag}(t_i)$, $\Lambda_i = \operatorname{diag}(\lambda_i)$, and the perturbed KKT residuals follow as

$$\begin{bmatrix} r_i^{\text{stat}} \\ r_i^{\text{eq}} \\ r_i^{\text{ineq}} \\ r_i^{\text{comp}} \end{bmatrix} := \begin{bmatrix} Hy_i + g + F^\top \pi_i + G^\top \lambda_i \\ Fy_i - b \\ Gy_i + t_i - d \\ T_i \Lambda_i \mathbf{1} - \tau_i \mathbf{1} \end{bmatrix}. \tag{2.7}$$

Eliminating Δt and $\Delta\lambda$ from the KKT system (2.6) by a Schur complement yields the symmetric saddle point system

$$\begin{bmatrix} \tilde{H}_i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta\pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}_i^{\text{stat}} \\ r_i^{\text{eq}} \end{bmatrix}, \tag{2.8}$$

with $\tilde{H}_i = H + G^\top T_i^{-1} \Lambda_i G \succeq H$ and $\tilde{r}_i^{\text{stat}} = r_i^{\text{stat}} + G^\top T_i^{-1} (\Lambda_i r_i^{\text{ineq}} - r_i^{\text{comp}})$. Notably, the term $G^\top T_i^{-1} \Lambda_i G$ can be regarded as the curvature contribution from

the inequality constraints. The weight on each constraint, i.e. the corresponding diagonal entry of $T_i^{-1}\Lambda_i$, increases as the constraint becomes active and tends to zero if it is inactive, and this barrier term is positive semidefinite, since t and λ are kept strictly positive throughout the iterations.

In the reduced KKT system of (2.8), assuming F has full row rank, the null space method can be applied to further reduce the system. Let the columns of Z form a basis of the null space of F, i.e. $FZ = 0$, and the columns of Y span the range of $F^\top$, so that $[\,Y \;\; Z\,]$ is nonsingular and FY is invertible.

The primal step can then be decomposed uniquely as $\Delta y = Y\,\Delta y_y + Z\,\Delta y_z$. Since $F\Delta y = FY\,\Delta y_y$, only the range-space component enters the equality constraints. This yields a system exposing the reduced Hessian as

$$\tilde{M}_i\,\Delta y_z = -\Big(Z^\top \tilde{r}_i^{\text{stat}} + Z^\top \tilde{H}_i\,Y\,\Delta y_y\Big), \tag{2.9}$$

where $Z^\top$ eliminates the equality Jacobian with $Z^\top F^\top = 0$, feasibility fixes the range-space component $\Delta y_y = -(FY)^{-1} r_i^{\text{eq}}$, and $\tilde{M}_i = Z^\top(H + G^\top T_i^{-1}\Lambda_i G)Z = Z^\top \tilde{H}_i Z$ is the reduced Hessian including the curvature contribution.

2.2 Efficient solution via Riccati recursion

In the context of optimal control, the decision variables y can be partitioned into state and control variables, and the QP in (2.1) takes the particular form

$$\begin{aligned}
\underset{\substack{x_0,\ldots,x_N\\ u_0,\ldots,u_{N-1}}}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left(\frac{1}{2} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \underbrace{\begin{bmatrix} Q_k & S_k^\top \\ S_k & R_k \end{bmatrix}}_{H_k} \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \begin{bmatrix} g_k^x \\ g_k^u \end{bmatrix}^\top \begin{bmatrix} x_k \\ u_k \end{bmatrix} \right) \\
& + \frac{1}{2} x_N^\top Q_N x_N + (g_N^x)^\top x_N \\
\text{subject to} \quad & x_0 = \hat{x}_0, \\
& x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, \ldots, N-1, \\
& G_k^x x_k + G_k^u u_k \le d_k, \qquad\quad k = 0, \ldots, N-1, \\
& G_N^x x_N \le d_N,
\end{aligned} \tag{2.10}$$

where $x_k \in \mathbb{R}^{n_x}$ and $u_k \in \mathbb{R}^{n_u}$ are the state and control at stage k, and N is the horizon length. The $Q_k \in \mathbb{R}^{n_x \times n_x}$, $R_k \in \mathbb{R}^{n_u \times n_u}$, and $S_k \in \mathbb{R}^{n_u \times n_x}$ are the state, control, and state–control coupling blocks of the full Hessian block H_k, $g_k^x \in \mathbb{R}^{n_x}$ and $g_k^u \in \mathbb{R}^{n_u}$ are the corresponding gradients, and Q_N, g_N^x are the terminal Hessian and gradient. The $A_k \in \mathbb{R}^{n_x \times n_x}$ and $B_k \in \mathbb{R}^{n_x \times n_u}$ are the Jacobians of the dynamics with respect to the state and control, $b_k \in \mathbb{R}^{n_x}$ is the dynamics residual, and $\hat{x}_0 \in \mathbb{R}^{n_x}$ is the initial state. The $G_k^x \in \mathbb{R}^{n_k^c \times n_x}$ and $G_k^u \in \mathbb{R}^{n_k^c \times n_u}$ are the Jacobians of the n_k^c inequality constraints at stage k, including linear constraints and box constraints on states and controls, with right-hand side $d_k \in \mathbb{R}^{n_k^c}$.

Following the same derivation as in Section 2.1, we can obtain the KKT system of (2.10), which exhibits a tridiagonal structure since the cost and inequality constraints are stage-wise and the dynamics couple only consecutive stages:

$$\begin{bmatrix} \ddots & & & & & \\ & 0 & -I & & & \\ & -I & \begin{pmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{pmatrix} & \begin{matrix} A_k^\top \\ B_k^\top \end{matrix} & & \\ & 0 & A_k \quad B_k & 0 & -I & \\ & & & -I & \ddots & \end{bmatrix} \begin{bmatrix} \vdots \\ \Delta\pi_k \\ \Delta x_k \\ \Delta u_k \\ \Delta\pi_{k+1} \\ \vdots \end{bmatrix} = - \begin{bmatrix} \vdots \\ r_{k-1}^{\text{eq}} \\ \tilde{r}_k^{\text{stat},x} \\ \tilde{r}_k^{\text{stat},u} \\ r_k^{\text{eq}} \\ \vdots \end{bmatrix}, \tag{2.11}$$

where the block matrices $\tilde{Q}_k$, $\tilde{R}_k$, and $\tilde{S}_k$ contain the curvature contributions from inequality constraints as

$$\begin{aligned} \tilde{Q}_k &= Q_k + (G_k^x)^\top \Lambda_k T_k^{-1} G_k^x, \\ \tilde{R}_k &= R_k + (G_k^u)^\top \Lambda_k T_k^{-1} G_k^u, \\ \tilde{S}_k &= S_k + (G_k^u)^\top \Lambda_k T_k^{-1} G_k^x. \end{aligned}$$

This tridiagonal structure allows for an efficient solution of the KKT system using the Riccati recursion [11], which consists of a backward and forward pass. The backward pass computes the feedback gains K_k and feedforward terms k_k, as well as the cost-to-go matrices P_k and vectors p_k, while the forward pass computes the optimal state and control trajectories. The algorithmic details of the Riccati recursion are summarized in Algorithm 2.

Algorithm 2 Riccati recursion for solving a QP with OCP structure in (2.10)

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}_N^{\mathrm{stat},x}$

// Backward recursion

for $k = N-1$ **down to** 0 **do**

$\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$

$\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

$K_k \leftarrow -\tilde{M}_k^{-1} \Gamma_k$

$k_k \leftarrow -\tilde{M}_k^{-1} \left(\tilde{r}_k^{\mathrm{stat},u} + B_k^\top (P_{k+1} r_k^{\mathrm{eq}} + p_{k+1}) \right)$

$P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k$

$p_k \leftarrow \tilde{r}_k^{\mathrm{stat},x} + A_k^\top (P_{k+1} r_k^{\mathrm{eq}} + p_{k+1}) + \Gamma_k^\top k_k$

end for

// Forward recursion

$\Delta x_0 \leftarrow \hat{x}_0 - x_0$

for $k = 0$ **to** $N-1$ **do**

$\Delta u_k \leftarrow K_k \Delta x_k + k_k$

$\Delta x_{k+1} \leftarrow A_k \Delta x_k + B_k \Delta u_k + r_k^{\mathrm{eq}}$

$\Delta \pi_{k+1} \leftarrow P_{k+1} \Delta x_{k+1} + p_{k+1}$

$\Delta t_k \leftarrow -(G_k^x \Delta x_k + G_k^u \Delta u_k + r_k^{\mathrm{ineq}})$

$\Delta \lambda_k \leftarrow T_k^{-1}(-r_k^{\mathrm{comp}} - \Lambda_k \Delta t_k)$

end for

$\Delta t_N \leftarrow -(G_N^x \Delta x_N + r_N^{\mathrm{ineq}})$

$\Delta \lambda_N \leftarrow T_N^{-1}(-r_N^{\mathrm{comp}} - \Lambda_N \Delta t_N)$

By applying the Riccati recursion, we solve the KKT system in (2.9) without factorizing the reduced Hessian $\tilde{M}_i$ as a whole, and the computational complexity is reduced from $\mathcal{O}(N^3)$ for a dense factorization to $\mathcal{O}(N)$ [11], thus achieving an efficient solution by exploiting the OCP structure.

2.3 Adaptive barrier update

In this section, two closely related adaptive barrier update schemes are discussed for the primal–dual IPM. While monotone updates of the barrier parameter τ, such as the Fiacco–McCormick scheme [12], enjoy strong convergence guarantees, they may lead to slow convergence and practical limitations [13]. We therefore prefer more aggressive, adaptive updates of τ for faster convergence, such as Mehrotra's predictor–corrector and Mehrotra's probing presented next. Both schemes set τ adaptively from an affine predictor step, and the predictor–corrector scheme additionally corrects the search direction. Section 2.3.3 then compares the two schemes for nonconvex QPs.

2.3.1 Mehrotra's predictor–corrector

For convex QPs, Mehrotra's predictor–corrector scheme [14], which adaptively updates the barrier parameter τ and adds a second-order corrector for improving the direction, is usually considered a good choice for fast convergence. In this scheme, the barrier parameter τ is defined as

$$\tau_i = \sigma_i \mu_i, \tag{2.12}$$

$$\mu_i = \frac{t_i^\top \lambda_i}{n_c}, \tag{2.13}$$

where μ_i is the duality measure, i.e. the average complementarity product, which measures the violation of the complementarity condition (2.4d), and $\sigma_i \geq 0$ is the centering parameter that controls the trade-off between centering and aggressive steps. In each iteration, two solves of (2.6) are performed that share the same left-hand side (LHS) and differ only in the complementarity at the right-hand side (RHS), and each iteration consists of a predictor (affine scaling) step, a centering

update, and a corrector step.

Predictor step. The scheme starts with computing a predictor step, also called the affine scaling step. For this step, the linear system in (2.6) is solved with $\sigma_i = 0$ in the complementarity residual r_i^{comp} and no correction considered, i.e. the resulting complementarity residual is written as

$$r_i^{\text{comp,aff}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1} = T_i \Lambda_i \mathbf{1}. \tag{2.14}$$

This yields the affine scaling direction $(\Delta y_{\text{aff}}, \Delta \pi_{\text{aff}}, \Delta \lambda_{\text{aff}}, \Delta t_{\text{aff}})$. The affine step is used to estimate the centering parameter σ_i and form the second-order correction in the following.

Centering update. After the predictor step is obtained, the centering parameter σ is updated based on the estimated duality measure μ_{aff} after the affine step is taken, which is defined as

$$\sigma_i = (\mu_{\text{aff}}/\mu_i)^3 \tag{2.15}$$

$$\mu_{\text{aff}} = \frac{(t_i + \alpha_i^{\text{prim,aff}} \Delta t_{\text{aff}})^\top (\lambda_i + \alpha_i^{\text{dual,aff}} \Delta \lambda_{\text{aff}})}{n_c}, \tag{2.16}$$

where the step sizes $\alpha_i^{\text{prim,aff}}$ and $\alpha_i^{\text{dual,aff}}$ are the largest step sizes that keep the slack variables t_i and Lagrange multipliers Λ_i nonnegative, i.e.

$$\alpha_i^{\text{prim,aff}} = \min \left(1, \min_{j:\Delta t_j^{\text{aff}} < 0} -\frac{t_{j,i}}{\Delta t_j^{\text{aff}}} \right), \tag{2.17}$$

$$\alpha_i^{\text{dual,aff}} = \min \left(1, \min_{j:\Delta \lambda_j^{\text{aff}} < 0} -\frac{\lambda_{j,i}}{\Delta \lambda_j^{\text{aff}}} \right). \tag{2.18}$$

If $\mu_{\text{aff}} \ll \mu_i$, the affine step achieves a substantial reduction in the duality gap, which drives σ_i sharply to 0, resulting in a near-Newton, aggressive step, whereas a stalled affine step with $\mu_{\text{aff}} \approx \mu_i$ gives $\sigma_i \to 1$ and thus a recentering step towards the central path.

Corrector step. In the corrector step, the linear system in (2.6) is solved again where the complementarity residual is modified to include the updated centering parameter σ_i and a second-order correction term using the calculated affine steps as follows:

$$r_i^{\text{comp,cor}} = T_i\Lambda_i\mathbf{1} + \Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1} - \sigma_i\mu_i\mathbf{1}, \tag{2.19}$$

where $\Delta T_{\text{aff}} = \text{diag}(\Delta t_{\text{aff}})$, $\Delta\Lambda_{\text{aff}} = \text{diag}(\Delta\lambda_{\text{aff}})$, the term $\Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1}$ cancels the second-order error that the affine step leaves in $t_j\lambda_j$, and $-\sigma_i\mu_i\mathbf{1}$ supplies the centering. This second solve returns the final direction $(\Delta y, \Delta\pi, \Delta\lambda, \Delta t)$ directly, and a step size control follows to ensure the positivity of the slack variables and Lagrange multipliers for the inequality constraints.

Condition on corrector step. In practice, the corrector step is usually applied conditionally. The mechanism is, therefore, called conditional Mehrotra's predictor–corrector scheme [13], where the corrector step is only accepted if it does not largely increase the duality measure after the corrector step is taken compared with that after the predictor step. More specifically, the duality measure after the corrector step is estimated as

$$\mu_{\text{cor}} = \frac{(t_i + \alpha_i^{\text{prim,cor}}\Delta t)^\top(\lambda_i + \alpha_i^{\text{dual,cor}}\Delta\lambda)}{n_c}, \tag{2.20}$$

where $\alpha_i^{\text{prim,cor}}$ and $\alpha_i^{\text{dual,cor}}$ are computed as the affine step sizes in (2.17), but along the corrected direction. If $\mu_{\text{cor}} \leq 2\mu_{\text{aff}}$, the corrected direction is accepted. Otherwise, the linear system in (2.6) is solved again without the correction term $\Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1}$ in (2.19).

2.3.2 Mehrotra's probing

In contrast to Mehrotra's predictor–corrector scheme or its conditional variant, Mehrotra's probing scheme [13] drops the corrector step entirely, resulting in a more conservative approach for updating the barrier parameter τ.

In Mehrotra's probing, after one solve of the linear system with (2.14) as the complementarity residual, another solve gives the so-called centering step, and

the corresponding complementarity residual at the RHS is given by

$$r_i^{\text{comp,pro}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1}, \tag{2.21}$$

where the centering parameter σ_i is updated using (2.15) after the affine step.

2.3.3 Comparison of schemes for nonconvex QPs

We argue that Mehrotra's predictor–corrector scheme is not necessarily the best choice in the nonconvex case. In the corrector step, $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ is added to the complementarity residual to include the complementarity error of a full affine step, which assumes the affine direction is an accurate Newton step. In nonconvex QPs, however, the Hessian at the LHS usually needs to be modified, and thus the direction can be adjusted for descent when the constraints are satisfied, so it is actually an inexact Newton step, violating the assumption of the predictor–corrector scheme.

Probing uses the affine step only through $\sigma = (\mu_{\text{aff}}/\mu)^3$, so an inexact affine step just gives $\sigma \approx 1$, i.e. a centering step. The iterate moves back to the central path instead of trying to decrease the incorrect complementarity error introduced by the inexact Newton step.

2.4 Step size control

In this section, we discuss the step size control employed in a primal–dual IPM. The fraction-to-the-boundary (FTB) rule is first introduced as a common strategy for step size control in IPMs, which ensures that the updated slack variables and Lagrange multipliers remain positive. Then, a two-type line search with backtracking for QPs is introduced, which is simplified from the funnel line search method introduced in [15] for the nonconvex case.

2.4.1 Fraction-to-the-boundary rule

The FTB rule is a common strategy for step size control in primal–dual IPMs. The maximal step sizes $\alpha_i^{\text{prim,max}}$ and $\alpha_i^{\text{dual,max}}$ are computed without violating the nonnegativity of the slack variables and Lagrange multipliers, as

$$\alpha_i^{\text{prim,max}} = \min_{j:\Delta t_j<0} -\frac{t_{j,i}}{\Delta t_j}, \tag{2.22}$$

$$\alpha_i^{\text{dual,max}} = \min_{j:\Delta \lambda_j<0} -\frac{\lambda_{j,i}}{\Delta \lambda_j}. \tag{2.23}$$

Then, the actual step sizes are chosen as a fraction of the maximum step sizes, i.e.,

$$\alpha_i^{\text{prim}} = \min(1, \eta\alpha_i^{\text{prim,max}}), \tag{2.24}$$

$$\alpha_i^{\text{dual}} = \min(1, \eta\alpha_i^{\text{dual,max}}), \tag{2.25}$$

where $\eta \in (0, 1)$ is a parameter that controls the fraction of the maximum step size to be taken, typically set to a value close to 1 to ensure sufficient progress while maintaining feasibility.

2.4.2 Two-type line search for quadratic programs

A line search with backtracking can further reduce the step size for a sufficient decrease of the primal infeasibility. In the nonconvex case, the regularized direction is not the exact Newton step, and the FTB step along a descent direction might still increase the barrier objective in an IPM. We simplify the funnel method proposed in [15] to a two-type line search tailored for QPs, and the two-type line search is executed after the FTB part is performed.

In an IPM for QPs, the objective ϕ used in the line search is the barrier objective

$$\phi(y, t) = \frac{1}{2}y^\top Hy + g^\top y - \tau \sum_{j=1}^{n_c} \ln t_j, \tag{2.26}$$

which is minimized subject to $Fy = b$ and $Gy + t = d$. With $\lambda_j = \tau/t_j$, the corresponding KKT conditions recover the relaxed complementarity in (2.5). The

line search consists of three parts: a switching condition, the f-type step and the h-type step.

The trial step is classified by the *switching condition*: it is an *f-type* step if

$$-\alpha_i^{\text{prim}} \rho_i > \kappa_{\text{s}}\, \theta_i^2, \qquad \kappa_{\text{s}} > 0, \tag{2.27}$$

holds, and an *h-type* step otherwise, where $\theta_i = \|(r_i^{\text{eq}}, r_i^{\text{ineq}})\|$ is the constraint violation at the current iterate and $\rho_i = \nabla\phi_i^\top \Delta z$ the directional derivative of ϕ along the primal search direction $\Delta z = (\Delta y, \Delta t)$. The condition depends only on the current iterate and the trial step size, so it is evaluated before a candidate is formed.

For an h-type step, the predicted objective decrease is small relative to the constraint violation. The FTB step is then accepted without evaluating the objective. The constraints of the QP are affine and the Newton direction solves the corresponding rows of the linear system exactly, so $\theta(\alpha_i^{\text{prim}}) = (1 - \alpha_i^{\text{prim}})\, \theta_i$ and the largest admissible step reduces the constraint violation the most.

For an f-type step, sufficient objective progress is expected. A candidate iterate $z^+(\alpha_i^{\text{prim}})$ is formed in the primal variables (y, t) and accepted if its objective ϕ^+ satisfies the Armijo condition [16]

$$\phi^+ \le \phi_i + \gamma\, \alpha_i^{\text{prim}}\, \rho_i, \qquad \gamma \in (0, 1). \tag{2.28}$$

Otherwise, the step size α_i^{prim} is reduced by $\beta \in (0, 1)$ and the search repeats. If all $n_{\max}$ trials are rejected, the FTB step size is restored. The dual step size α_i^{dual} is not changed. The procedure is summarized in Algorithm 3.

Two simplifications distinguish this procedure from the funnel method of [15]. First, no funnel bound $\theta_{\max}$ is maintained. In the funnel method, a trial iterate is admissible only if $\theta(z) \le \theta_{\max}$, and the bound $\theta_{\max}$ is decreased whenever an h-step is accepted. Here $\theta(\alpha_i^{\text{prim}}) = (1 - \alpha_i^{\text{prim}})\, \theta_i \le \theta_i$ for every $\alpha_i^{\text{prim}} \in (0, 1]$, so no trial step increases the constraint violation, and the funnel itself can then be removed. Second, since H is generally indefinite, $\Delta y^\top H \Delta y$ can have either sign, and the second-order term is unreliable for classifying the step. Instead of considering the second-order model as in [15], the switching condition in (2.27) uses only the first-order term $-\rho_i$, as the Armijo condition in (2.28) does. To

Algorithm 3 Two-type line search

Input: iterate z_i, direction Δz, FTB steps $\alpha_i^{\text{prim}}, \alpha_i^{\text{dual}}$, ϕ_i, θ_i, ρ_i
Parameters: $\kappa_{\text{s}} > 0$, $\gamma, \beta \in (0,1)$, $n_{\max} \in \mathbb{N}$
$\alpha_i^{\text{prim,FTB}} \leftarrow \alpha_i^{\text{prim}}$
for $\text{ls} = 1$ **to** $n_{\max}$ **do**
  if $\alpha_i^{\text{prim}}(-\rho_i) \leq \kappa_{\text{s}}\,\theta_i^2$ **then** ▷ h-type: feasibility dominates
    break ▷ accept FTB-scaled step
  end if
  form candidate $z^+(\alpha_i^{\text{prim}})$
  compute ϕ^+
  if $\phi^+ \leq \phi_i + \gamma\,\alpha_i^{\text{prim}}\,\rho_i$ **then** ▷ f-type: Armijo
    break
  end if
  if $\text{ls} = n_{\max}$ **then**
    $\alpha_i^{\text{prim}} \leftarrow \alpha_i^{\text{prim,FTB}}$ ▷ all trials rejected: revert to the FTB step
  else
    $\alpha_i^{\text{prim}} \leftarrow \beta\,\alpha_i^{\text{prim}}$
  end if
end for
return $\alpha_i^{\text{prim}},\ \alpha_i^{\text{dual}}$

compensate for this, κ_{s} is set small, so that objective-reducing steps are not classified as h-type steps.

Chapter 3

Primal regularization for indefinite quadratic programs

In this chapter, we discuss the treatment of QPs with indefinite full Hessian matrices, referred to as indefinite QPs. We first review numerical strategies to regularize the Hessian matrix, and motivate the reduced Hessian as the target for the detection and the regularization in Section 3.1. In Section 3.2, we propose Riccati-based primal regularizations, which detect the indefiniteness and apply the regularization inside the recursion. Next, we analyze in Section 3.3 why the minimum perturbation rule fails in this setting, and derive an improved stage-wise regularization strategy from that analysis. Finally, in Section 3.4, we prove that, owing to the detection on the reduced Hessian, Riccati-based regularizations become inactive as the iterates approach the solution.

3.1 Primal regularization

In indefinite QPs, a descent search direction in the IPM cannot be ensured when the constraints are satisfied, so regularizations need to be applied to the Hessian matrix to ensure such a direction. The resulting regularized KKT system after the Schur complement can be generally presented as

$$\begin{bmatrix} \tilde{H}_i + E_i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}_i^{\mathrm{stat}} \\ r_i^{\mathrm{eq}} \end{bmatrix}, \tag{3.1}$$

where E_i is an adjustment matrix that renders the adjusted Hessian $\tilde{H}_i + E_i$ positive definite at the i-th iteration. This process is referred to as primal regularization [17].

We start with the numerical strategies to compute E_i in Section 3.1.1, then discuss the detection of the indefiniteness in Section 3.1.2, and finally the regularization on the full or the reduced Hessian in Section 3.1.3.

3.1.1 Regularization strategies

In order to compute the adjustment matrix E_i, three categories of strategies are commonly used, which are eigenvalue modification, modified Cholesky factorization and Levenberg–Marquardt [3, 18, 19]. From these categories we select four methods to be implemented and compared: project and mirror [9] from eigenvalue modification, Gill–Murray–Wright (GMW) [20] from modified Cholesky factorization, and inertia correction [17] from Levenberg–Marquardt.

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix. A admits the eigenvalue decomposition $A = VD_EV^\top$, where V is orthogonal and D_E is the diagonal matrix of eigenvalues. If A is positive definite, it also admits the modified Cholesky factorization $A = LD_IL^\top$, with L unit lower triangular and D_I diagonal. In this notation, the four methods read as follows:

Eigenvalue modification: project and mirror. For both project and mirror, the eigenvalue decomposition $A = VD_EV^\top$ is first computed, and the eigenvalues in D_E are then modified in different manners,

$$\begin{aligned}\text{project}(A,\epsilon) &:= V\max(\epsilon I, D_E)V^\top,\\ \text{mirror}(A,\epsilon) &:= V\max(\epsilon I, |D_E|)V^\top,\end{aligned}$$

where $\epsilon > 0$ is a small constant and the max operator is applied component-wise. Project clips any negative or near-zero eigenvalues to ϵ, which keeps a minimal adjustment to the original matrix for positive definiteness, thus achieving the minimal Frobenius norm of the adjustment matrix E [3], but may lead to an increase in the condition number if the original eigenvalues were not close to zero.

Mirror reflects negative eigenvalues to their positive counterparts. It preserves the magnitude of the curvature in all directions, which avoids the ill-conditioning of project at the cost of a larger perturbation when $|\nu_{\min}| > \epsilon$.

Modified Cholesky factorization: GMW. A modified Cholesky factorization determines E during the factorization itself, with four objectives [21]. First, E should be zero if A is sufficiently positive definite. Second, if A is indefinite, $\|E\|$ should not be much larger than $\inf\{\|\Delta A\| : A + \Delta A \succeq 0\}$ for some reasonable norm. Third, $A + E$ should be reasonably well-conditioned. Fourth, since the standard Cholesky factorization of A costs about $\frac{1}{3}n^3$ flops, the modification should add only a small multiple of n^2 operations, so that the cost remains that of one Cholesky factorization to leading order.

According to how they correct the pivots D_I, similarly to how mirror and project correct eigenvalues, modified Cholesky factorizations fall into two types [21]. A Type-I strategy sets $\hat{D}_I = \max(|D_I|, \epsilon)$, whereas a Type-II strategy sets $\hat{D}_I = \max(D_I, \epsilon)$. Type-I thus corresponds to mirror and Type-II to project, so Type-I favors a well-conditioned $A + E$ and Type-II a small $\|E\|$.

GMW is the most common Type-I strategy and, unlike the Type-II strategies of Schnabel–Eskow [22] and Cheng–Higham [23], requires neither Gerschgorin bounds nor block pivoting [18]. It directly performs the $\mathrm{LDL}^\top$ factorization and corrects its pivots, whose signs give the inertia of A [20]. During the GMW factorization, each pivot is replaced by the largest of three values: its absolute value which flips the value if negative and remains the same if positive, a small value ϵ for lifting a near-zero pivot, and the smallest value for keeping the entries of the corresponding column of $LD_I^{1/2}$ below a fixed bound. The first two values form the Type-I rule $\max(|D_I|, \epsilon)$, and the third only adds the bound on L for numerical stability.

The increments form a nonnegative diagonal correction E with $A + E = LD_IL^\top$, which avoids the cost of an eigendecomposition and still guarantees positive definiteness, but E is generally not minimal and tends to over-modify A.

Levenberg-Marquardt: inertia correction. In this strategy, a uniform multiple of the identity is instead added to shift every eigenvalue by the same amount.

The method is easy to implement, but, because it acts equally in all directions, a shift large enough for a few negative eigenvalues also distorts the eigenvalues that were already positive. The operation can be described as

$$\mathrm{ic}(A, \delta_w) := A + \delta_w I, \qquad \delta_w \geq 0.$$

In practice, δ_w needs to be found by repeatedly increasing δ_w until the Cholesky factorization of $A + \delta_w I$ is successful. This strategy is widely used in IPOPT [17], Fatrop [24] and Crocoddyl [25].

3.1.2 Indefiniteness detection on reduced Hessian

Though the positive definiteness of the full Hessian H is rather easy to detect and is sufficient for convexity, an equality-constrained QP only requires the reduced Hessian to be positive definite for a unique global minimum. The following theorem, stated in [9] (for a proof, see [3]), makes this precise:

Theorem 1. *Assume that linear independence constraint qualification (LICQ) holds. A QP with only equality constraints has a unique global optimum if and only if the reduced Hessian is positive definite:*

$$M = Z^\top H Z \succ 0, \tag{3.2}$$

where the columns of Z span the null space of the equality constraint Jacobian F. The positive definiteness of the reduced Hessian is thus a necessary and sufficient condition for the equality-constrained QP admitting a unique global optimum.

Since the KKT system after the Schur complement in (2.8) is that of an equality-constrained QP as

$$\begin{aligned} \underset{\Delta y}{\text{minimize}} \quad & \frac{1}{2}\Delta y^\top \tilde{H}_i \Delta y + (\tilde{r}_i^{\text{stat}})^\top \Delta y \\ \text{subject to} \quad & F\Delta y + r_i^{\text{eq}} = 0, \end{aligned} \tag{3.3}$$

by Theorem 1, the step Δy is the unique minimizer of this QP if and only if $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$.

Based on this, a detection of the indefiniteness of $\tilde{M}_i$ is therefore preferable to that of $\tilde{H}_i$, which avoids unnecessary regularization of $\tilde{H}_i$ when $\tilde{H}_i$ is indefinite but $\tilde{M}_i \succ 0$.

3.1.3 Regularization on full or reduced Hessian

Once indefiniteness is detected, regularizations should be applied to the Hessian so that Δy can be a descent direction when the constraints are satisfied. Either the full or the reduced Hessian can be the target matrix, which the correction can be derived from and applied to.

The full Hessian regularization takes the full Hessian $\tilde{H}_i$ as the target. In optimal control, it is applied stage-wise to the block $\tilde{H}_k$ of each stage k to preserve the OCP structure,

$$\tilde{H}_{k,i}^{\mathrm{reg}} = \begin{bmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{bmatrix} + E_k \succ 0, \quad k = 0, 1, \ldots, N-1. \tag{3.4}$$

As each E_k depends only on $\tilde{H}_k$, this is simple to implement. It ensures positive curvature in every direction, i.e. $\Delta y^\top \tilde{H}_i^{\mathrm{reg}} \Delta y > 0$ also for $\Delta y = Y \Delta y_y + Z \Delta y_z$ with $\Delta y_y \neq 0$, which matters at infeasible iterates where Δy leaves the null space of F. However, it could be conservative when equality constraints are satisfied. At an iterate with $r_i^{\mathrm{eq}} = 0$, Δy lies in the null space of F, so only the curvature of $\tilde{H}_i$ within this subspace affects the step, and removing negative curvature outside it makes the adjustment larger than necessary.

The following result shows that, with a part adapted from [3], when equality constraints are satisfied, positive curvature in the null space is both necessary and sufficient for a descent direction.

Theorem 2. *Assume that LICQ holds, $r_i^{\mathrm{eq}} = 0$, and $\tilde{M}_i = Z^\top \tilde{H}_i Z$ is nonsingular. Then $(\tilde{r}_i^{\mathrm{stat}})^\top \Delta y < 0$ holds for every $\tilde{r}_i^{\mathrm{stat}}$ with $Z^\top \tilde{r}_i^{\mathrm{stat}} \neq 0$ if and only if $\tilde{M}_i \succ 0$.*

Proof. Since $r_i^{\mathrm{eq}} = 0$, the range-space component $\Delta y_y = -(FY)^{-1} r_i^{\mathrm{eq}}$ vanishes, so

$\Delta y = Z\Delta y_z$ and (2.9) reduces to $\tilde{M}_i \Delta y_z = -Z^\top \tilde{r}_i^{\text{stat}}$. Hence

$$(\tilde{r}_i^{\text{stat}})^\top \Delta y = -(Z^\top \tilde{r}_i^{\text{stat}})^\top \tilde{M}_i^{-1} (Z^\top \tilde{r}_i^{\text{stat}}). \tag{3.5}$$

For sufficiency, if $\tilde{M}_i \succ 0$, then $\tilde{M}_i^{-1} \succ 0$, and the RHS of (3.5) is negative for every $Z^\top \tilde{r}_i^{\text{stat}} \neq 0$.

For necessity, we proceed by contraposition. Suppose that $\tilde{M}_i$ is not positive definite. Being nonsingular, it has an eigenpair (ν, v) with $\nu < 0$. Since Z has full column rank, $\tilde{r}_i^{\text{stat}} = Z(Z^\top Z)^{-1} v$ satisfies $Z^\top \tilde{r}_i^{\text{stat}} = v \neq 0$, and (3.5) with $\tilde{M}_i^{-1} v = v/\nu$ gives $(\tilde{r}_i^{\text{stat}})^\top \Delta y = -v^\top v/\nu > 0$, so Δy is an ascent direction for this residual. □

At feasible iterates, a positive definite reduced Hessian is therefore exactly what a descent direction requires, and the reduced Hessian regularization, which takes $\tilde{M}_i$ as the target, enforces this condition.

However, $\tilde{M}_i$ is usually not directly available. Computing it requires a null-space basis of the equality constraints, e.g. via condensing in optimal control [26], which not only introduces extra computational cost but also yields a dense $\tilde{M}_i$, losing the stage-wise structure preserved by the full Hessian regularization. When both the detection and the regularization operate on this dense matrix, the factorization cost grows cubically with the horizon N instead of linearly as in the Riccati recursion, which contradicts the structure-exploiting purpose.

3.2 Riccati-based primal regularization

The discussion in the previous section motivates the investigation of Riccati-based primal regularizations, which aim at regularizing the reduced Hessian during the Riccati recursion so that the OCP structure is preserved. In Riccati-based regularizations, the Riccati recursion serves two purposes. First, it detects the indefiniteness of the reduced Hessian $\tilde{M}_i$ through the failure of a Cholesky factorization, as shown in Section 3.2.1. Second, it regularizes the reduced Hessian $\tilde{M}_i$ stage-wise to ensure a positive definite reduced Hessian $\tilde{M}_i$ while preserving the OCP structure, as described in Section 3.2.2.

3.2.1 Reduced Hessian blocks from Riccati recursion

In this subsection, we restate and establish the connection between the Riccati recursion and the block $\mathrm{LDL}^\top$ factorization of the reduced Hessian $\tilde{M}_i$, which provides a way to implicitly compute $\tilde{M}_i$ while simultaneously detecting its indefiniteness.

Algorithm 4 Backward pass of Riccati recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}_N^{\mathrm{stat},x}$

for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $L_k \leftarrow \mathrm{Cholesky}(\tilde{M}_k)$
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}_k^{\mathrm{stat},u} + B_k^\top (P_{k+1} r_k^{\mathrm{eq}} + p_{k+1})\right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}_k^{\mathrm{stat},x} + A_k^\top (P_{k+1} r_k^{\mathrm{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

In practice, the backward pass of the Riccati recursion is implemented as in Algorithm 4, where $\tilde{M}_k$ is factorized by a Cholesky factorization and K_k, k_k follow from triangular solves, which is more efficient and succeeds only if $\tilde{M}_k \succ 0$.

Recall that the second-order term of (3.3) is stage-wise in optimal control:

$$\frac{1}{2} \sum_{k=0}^{N-1} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix}^\top \begin{bmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} + \frac{1}{2} \Delta x_N^\top \tilde{Q}_N \Delta x_N.$$

Condensing eliminates the states through the dynamics $\Delta x_{k+1} = A_k \Delta x_k + B_k \Delta u_k + r_k^{\mathrm{eq}}$ and permutes the dense reduced Hessian $\tilde{M}_i$ in $\Delta u_0, \ldots, \Delta u_{N-1}$. Since Δu_{N-1} enters only the last stage term and Δx_N, its diagonal block is

$$(\tilde{M}_i)_{N-1,N-1} = \tilde{R}_{N-1} + B_{N-1}^\top \tilde{Q}_N B_{N-1},$$

which is exactly $\tilde{M}_{N-1}$ in Algorithm 4 with $P_N = \tilde{Q}_N$. After eliminating Δu_{N-1}, the Schur complement is the reduced Hessian of the problem on stages $0, \ldots, N-2$ with terminal weight P_{N-1}, which has the same structure. Repeating this

backward in time to Δu_0 gives the block $\mathrm{LDL}^\top$ factorization

$$\tilde{M}_i = L \begin{bmatrix} \tilde{M}_{N-1}^i & 0 & \cdots & 0 \\ 0 & \tilde{M}_{N-2}^i & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{M}_0^i \end{bmatrix} L^\top$$

where L is unit lower triangular, with off-diagonal blocks built from the feedback gains K_k and the dynamics.

This confirms that the matrices $\tilde{M}_k$ from the Riccati recursion are the diagonal blocks of the block $\mathrm{LDL}^\top$ factorization of $\tilde{M}_i$. By Sylvester's law of inertia, $\tilde{M}_i \succ 0$ if and only if every $\tilde{M}_k^i \succ 0$, which indicates the Cholesky factorizations of all $\tilde{M}_k$ in the backward pass succeed. We can, therefore, use the Riccati recursion to detect indefiniteness, as also used in Fatrop [24].

3.2.2 Stage-wise regularization in Riccati recursion

With this connection, we can regularize $\tilde{M}_i$ stage-wise during the Riccati recursion, which preserves the sparse structure of the KKT system. Whenever the Cholesky factorization of a block $\tilde{M}_k$ fails, we add a block adjustment matrix E_k to it on the fly, i.e. use $\tilde{M}_k + E_k$. This modifies the backward pass as in Algorithm 5.

The regularization is thus referred to as Riccati-based regularization, and block adjustment matrices E_k can be derived by any of the strategies mentioned in Section 3.1.1.

3.3 Improved stage-wise regularization algorithm

Section 3.3.1 analyzes the case where the commonly used minimum perturbation strategy fails in stage-wise regularization, which would severely ill-condition the preceding stages due to the propagation of the small curvature. This motivates the improved stage-wise regularization in Section 3.3.2, which avoids minimum perturbation and accounts for the propagation.

Algorithm 5 Backward pass of modified Riccati recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}_N^{\text{stat},x}$, $\mathcal{E}(\cdot)$: strategy from Section 3.1.1

for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  if factorization fails **then**
    $E_k \leftarrow \mathcal{E}(\tilde{M}_k) - \tilde{M}_k$ ▷ Compute block adjustment
    $\tilde{M}_k \leftarrow \tilde{M}_k + E_k$ ▷ Apply regularization
    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  end if
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}_k^{\text{stat},u} + B_k^\top (P_{k+1} r_k^{\text{eq}} + p_{k+1}) \right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}_k^{\text{stat},x} + A_k^\top (P_{k+1} r_k^{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

3.3.1 Failure of minimum perturbation regularization

In this subsection, we show with a general example that the minimum perturbation strategy, e.g. project, which is commonly the first choice because it modifies the original problem the least, can cause a condition number issue in the stage-wise regularization.

Suppose at stage k, the reduced Hessian block $\tilde{M}_k$ is not positive definite. Let us assume $\tilde{M}_k$ has only one negative eigenvalue ν_j, and the corresponding eigenvector is v_j. The minimum perturbation strategy is applied as

$$E_k = (-\nu_j + \epsilon) v_j v_j^\top, \tag{3.6}$$

where ϵ is a small positive value to ensure the positive definiteness of the regularized reduced Hessian block.

By the Sherman–Morrison formula, the inverse of the regularized reduced Hessian

block can be expressed as

$$\begin{aligned}(\tilde{M}_k + E_k)^{-1} &= \tilde{M}_k^{-1} + \frac{\nu_j - \epsilon}{\epsilon\nu_j} v_j v_j^\top \\ &= \tilde{M}_k^{-1} + \left(\frac{1}{\epsilon} - \frac{1}{\nu_j}\right) v_j v_j^\top,\end{aligned}$$

and the update of the cost-to-go matrix P_k follows as

$$\begin{aligned}P_k &= \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k \\ &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k - \left(\frac{1}{\epsilon} - \frac{1}{\nu_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k \\ &= P_k^{\text{unreg}} - \left(\frac{1}{\epsilon} - \frac{1}{\nu_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k. \end{aligned} \tag{3.7}$$

Note that this negative rank-one modification can already render P_k itself indefinite, independently of its downstream effect on $\tilde{M}_{k-1}$. This can happen even when the perturbation on $\tilde{M}_k$ is moderate, provided that the negative-curvature direction v_j is strongly coupled to the state, i.e. $\|\Gamma_k^\top v_j\|$ is large.

Consequently, the reduced Hessian block $\tilde{M}_{k-1}$ at the previous stage $k-1$ is calculated as

$$\tilde{M}_{k-1} = \tilde{M}_{k-1}^{\text{unreg}} - \left(\frac{1}{\epsilon} - \frac{1}{\nu_j}\right) (B_{k-1}^\top \Gamma_k^\top v_j)(B_{k-1}^\top \Gamma_k^\top v_j)^\top, \tag{3.8}$$

and the condition number of the regularized reduced Hessian block $\tilde{M}_{k-1}$ can be approximated as

$$\kappa(\tilde{M}_{k-1}) \approx \frac{c(\epsilon + |\nu_j|)}{\epsilon |\nu_j|}, \qquad c = \frac{\|B_{k-1}^\top \Gamma_k^\top v_j\|^2}{\nu_{\min}(\tilde{M}_{k-1}^{\text{unreg}})}, \tag{3.9}$$

where c is an $O(1)$ constant. As ϵ is a small positive value, the presence of the negative rank-one update, scaled by roughly $\frac{1}{\epsilon}$, dominates, deteriorating the condition number of $\tilde{M}_{k-1}$, and also likely making it indefinite, which can further cause the need for successively larger regularizations through the stages and numerical explosion in the end.

3.3.2 Insights and improved stage-wise regularization

The issue in Section 3.3.1 reveals two insights into stage-wise regularization:

1. A larger perturbation norm needs to be considered for a reasonable condition number, similar to the discussion in [20].
2. The cost-to-go matrices P_k need to be incorporated into the regularization design, as a purely local scheme cannot capture the severity of propagation.

From the first insight, instead of the minimum perturbation strategy, an intuitive, but effective, approach is to use the absolute value of the negative eigenvalues or $\mathrm{LDL}^\top$ pivots as the perturbation magnitude, which can avoid the severe condition number issue in the preceding stages. This corresponds to mirror in eigenvalue modification and to the Type-I strategy, of which GMW is a common representative, in modified Cholesky factorization. From the second insight, we can incorporate the cost-to-go matrices P_k into the algorithm, so that the propagation of the small curvature can be taken into account in the regularization. The resulting improved stage-wise regularization algorithm is summarized in Algorithm 6.

Algorithm 6 Backward pass of improved modified Riccati recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}_N^{\mathrm{stat},x}$, $\mathcal{E}(\cdot)$: mirror or Type-I strategy

for $k = N-1$ **down to** 0 **do**
  $P_{k+1}^{\mathrm{reg}} \leftarrow P_{k+1}$ ▷ Default with no regularization
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1}^{\mathrm{reg}} B_k$
  $L_k \leftarrow \mathrm{Cholesky}(\tilde{M}_k)$
  if factorization fails **then**
    $P_{k+1}^{\mathrm{reg}} \leftarrow \mathcal{E}(P_{k+1})$ ▷ Regularize the cost-to-go matrix
    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1}^{\mathrm{reg}} B_k$
    $\tilde{M}_k^{\mathrm{reg}} \leftarrow \mathcal{E}(\tilde{M}_k)$ ▷ Regularize the reduced Hessian block
    $L_k \leftarrow \mathrm{Cholesky}(\tilde{M}_k^{\mathrm{reg}})$
  end if
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1}^{\mathrm{reg}} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}_k^{\mathrm{stat},u} + B_k^\top (P_{k+1}^{\mathrm{reg}} r_k^{\mathrm{eq}} + p_{k+1}) \right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1}^{\mathrm{reg}} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}_k^{\mathrm{stat},x} + A_k^\top (P_{k+1}^{\mathrm{reg}} r_k^{\mathrm{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

In Algorithm 6, both corrections on the reduced Hessian blocks $\tilde{M}_k$ and cost-to-go matrices P_k are produced by the same non-minimal strategy $\mathcal{E}(\cdot)$, mirror or Type-I. This controls the condition number of the regularized blocks, as motivated by the first insight, and takes the propagation into account, as in the second insight. When no factorization failure occurs, the recursion reduces to the standard Riccati backward pass.

3.4 Asymptotic inactivity of the regularization

Since the regularization is applied only as the detection triggers it, its behavior close to the solution is governed by the reduced Hessian $\tilde{M}_i$. In this section, we prove that any regularization triggered by this detection is inactive near the solution, i.e., it vanishes once the iterates are sufficiently close to the local minimizer. The argument adapts that of [27] to the reduced Hessian $\tilde{M}_i$.

Let $\mathcal{A} = \{j \mid t_j^* = 0\}$ denote the active set at y^*, split into the strongly active set $\mathcal{A}_+ = \{j \in \mathcal{A} \mid \lambda_j^* > 0\}$ and the weakly active set $\mathcal{A}_0 = \{j \in \mathcal{A} \mid \lambda_j^* = 0\}$. The critical cone [3] is defined as:

$$C(y^*, \lambda^*) = \{w \mid Fw = 0,\ G_j w = 0 \text{ for } j \in \mathcal{A}_+,\ G_j w \leq 0 \text{ for } j \in \mathcal{A}_0\},$$

where F and G are the equality and inequality constraint Jacobians and G_j denotes the j-th row of G.

For convenience, we restate the system in (2.9) at iteration i, which is used for the proof,

$$\tilde{M}_i \, \Delta y_z = -\Big(Z^\top \tilde{r}_i^{\text{stat}} + Z^\top \tilde{H}_i \, Y \, \Delta y_y\Big),$$

with the reduced Hessian $\tilde{M}_i = Z^\top \tilde{H}_i Z$ and $\tilde{H}_i = H + G^\top T_i^{-1} \Lambda_i G$, where the columns of Z span the null space of F, and T_i and Λ_i are the diagonal matrices of t_i and λ_i. We make the following assumptions for the proof:

Assumption 1. *(y^*, λ^*) is a local minimizer for the QP in (2.2).*

Assumption 2. *Each local minimizer (y^*, λ^*) for the QP in (2.2) is isolated.*

Assumption 3. *LICQ holds at the minimizer* (y^*, λ^*).

Assumption 4. *Strict complementarity holds at the minimizer* (y^*, λ^*).

Assumptions 1–3 imply the second-order sufficient condition (SOSC) for local optimality, so we have the following Lemma 3:

Lemma 3. *Under Assumptions 1–3, for all* $w \in C(y^*, \lambda^*) \setminus \{0\}$*, it holds that* $w^\top H w > 0$.

Proof. With Assumption 1, the second-order necessary condition (SONC) for local optimality holds at (y^*, λ^*), which states that for all $w \in C(y^*, \lambda^*)$, we have $w^\top H w \geq 0$.

We proceed by contradiction. Suppose there exists $w \in C(y^*, \lambda^*) \setminus \{0\}$ such that $w^\top H w = 0$. Consider the point $y(\alpha) = y^* + \alpha w$ for $\alpha > 0$. Because the constraints are linear and w is in the critical cone, $y(\alpha)$ remains feasible for sufficiently small α.

Since the objective function of (2.2) is quadratic, expanding $f(y(\alpha))$ yields

$$f(y(\alpha)) = f(y^*) + \alpha \nabla f(y^*)^\top w + \frac{1}{2}\alpha^2 w^\top H w$$

As $\nabla f(y^*)^\top w = 0$, given the assumption that $w^\top H w = 0$, the equation reduces exactly to $f(y(\alpha)) = f(y^*)$. This implies that $y(\alpha)$ is also a local minimizer, which directly contradicts Assumption 2. Therefore, $w^\top H w > 0$ must hold. □

Proposition 4. *Let Assumptions 1–4 hold, and let the regularization be triggered by the Cholesky factorization failure in Riccati recursion. If the iterates* (y_i, λ_i) *converge to* (y^*, λ^*)*, then for all sufficiently large* i *the reduced Hessian* $\tilde{M}_i$ *is positive definite, and hence no regularization is applied, i.e.* $E_i = 0$.

Proof. Under Assumption 4, $\mathcal{A}_0 = \emptyset$ and $\mathcal{A} = \mathcal{A}_+$, so the critical cone $C(y^*, \lambda^*)$ reduces to the null space of the active constraints. Therefore, we have $C(y^*, \lambda^*) = \{w \mid Fw = 0,\ G_{\mathcal{A}} w = 0\}$.

For any v with $\|v\| = 1$, write $w = Zv \in \{w \mid Fw = 0\} \setminus \{0\}$ and let G_j denote the j-th row of G. The corresponding quadratic form of the reduced Hessian reads

$$\begin{aligned} v^\top \tilde{M}_i v &= w^\top (H + G^\top T_i^{-1} \Lambda_i G) w \\ &= w^\top H w + \sum_j \frac{\lambda_{j,i}}{t_{j,i}} (G_j w)^2 \\ &= w^\top H w + \sum_{j \in \mathcal{A}} \frac{\lambda_{j,i}}{t_{j,i}} (G_j w)^2 + \sum_{j \notin \mathcal{A}} \frac{\lambda_{j,i}}{t_{j,i}} (G_j w)^2. \end{aligned}$$

We now proceed by contradiction. Assume that $\tilde{M}_i$ is not positive definite for infinitely many i. Then, passing to a subsequence, there exist v_i with $\|v_i\| = 1$ and $v_i \to \bar{v}$ such that $v_i^\top \tilde{M}_i v_i \leq 0$ for all i in this subsequence. Writing $w_i = Zv_i$ and $\bar{w} = Z\bar{v}$, so that $w_i \to \bar{w}$, we can distinguish two cases:

For $G_{\mathcal{A}} \bar{w} = 0$, the limit direction satisfies $\bar{w} \in C(y^*, \lambda^*) \setminus \{0\}$, so $\bar{w}^\top H \bar{w} > 0$ follows by Lemma 3. Since the remaining terms on the RHS are nonnegative, $v_i^\top \tilde{M}_i v_i \geq w_i^\top H w_i$ with $w_i^\top H w_i \to \bar{w}^\top H \bar{w} > 0$. Hence $v_i^\top \tilde{M}_i v_i > 0$ for all sufficiently large i, contradicting the assumption.

For $G_{\mathcal{A}} \bar{w} \neq 0$, Assumption 4 gives $\lambda_{j,i} \to \lambda_j^* > 0$ and $t_{j,i} \to 0$ for $j \in \mathcal{A}$, so the term $\sum_{j \in \mathcal{A}} \frac{\lambda_{j,i}}{t_{j,i}} (G_j w_i)^2 \to \infty$, which dominates the other two terms, and thus $v_i^\top \tilde{M}_i v_i \to \infty$, contradicting the assumption.

Therefore, we can conclude that $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$ for all i sufficiently large. By Section 3.2.1, all Cholesky factorizations in the backward pass then succeed and $E_i = 0$. □

Proposition 4 shows that once the iterates are close to (y^*, λ^*), $E_i = 0$ and the regularized recursion coincides with the unregularized one. The solver thus recovers the exact Newton step and inherits the local convergence rate of the underlying IPM.

Chapter 4

A benchmark for indefinite OCP QPs

The goal of this chapter is to provide test problems for evaluating the performance of the proposed Riccati-based primal regularizations on the subproblems of SQP. To this end, we construct a benchmark of indefinite OCP QPs that arise in the SQP iterations of NMPC problems. First, in Section 4.1, we introduce the construction of the QP benchmark, including the formulation of QP subproblems in the benchmark, an overview of OCP test problems and the pipeline for collecting indefinite QPs. Then we analyze the collected problem instances to gain a better understanding of characteristics regarding definiteness in the indefinite OCP QP benchmark in Section 4.2.

4.1 Benchmark construction

This section describes the construction of the indefinite QP benchmark. We first formulate the QP subproblems arising in SQP iterations of the OCPs, then introduce the eight OCP test problems, formulated as fixed-time regulator problems with quadratic stage costs and box constraints on the controls, and finally the collection pipeline.

4.1.1 Formulation of QP subproblems from OCPs

We start from the formulation of the nonlinear OCPs to obtain the resulting QP subproblems that arise in the SQP iterations. In the benchmark, all OCPs share the same structure: a fixed-horizon, direct multiple-shooting OCP with a quadratic stage cost that penalizes deviations from given state and control references. The mathematical formulation of the OCPs is given as follows:

$$\begin{aligned}
\underset{x,\,u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \frac{1}{2} \begin{bmatrix} x_k - x_k^{\text{ref}} \\ u_k - u_k^{\text{ref}} \end{bmatrix}^\top \begin{bmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{bmatrix} \begin{bmatrix} x_k - x_k^{\text{ref}} \\ u_k - u_k^{\text{ref}} \end{bmatrix} \\
& + \tfrac{1}{2}(x_N - x_N^{\text{ref}})^\top \mathcal{Q}_N (x_N - x_N^{\text{ref}}) \\
\text{subject to} \quad & x_0 = \bar{x}_0, \\
& x_{k+1} = f_k(x_k, u_k), && k = 0, \dots, N-1, \\
& \underline{g} \le \mathcal{C}_k x_k + \mathcal{D}_k u_k \le \bar{g}, && k = 0, \dots, N-1, \\
& \underline{h} \le h_k(x_k, u_k) \le \bar{h}, && k = 0, \dots, N-1, \\
& \underline{h}_N \le h_N(x_N) \le \bar{h}_N,
\end{aligned} \tag{4.1}$$

where $x_{k+1} = f_k(x_k, u_k)$ is one integration step of the continuous dynamics over $[t_k, t_{k+1}]$. The given initial state is $\bar{x}_0$, $\mathcal{Q}_k, \mathcal{R}_k, \mathcal{Q}_N \succeq 0$ are weighting matrices, $\mathcal{S}_k$ is the coupling cost matrix, and $x_k^{\text{ref}}, u_k^{\text{ref}}$ are the reference points. The term $\mathcal{C}_k x_k + \mathcal{D}_k u_k$ includes the affine and box path constraints, and $h_k(\cdot,\cdot)$ and $h_N(\cdot)$ collect the nonlinear path and terminal constraints, respectively.

When using the exact Hessian of the Lagrangian in the SQP iterations, the resulting Hessian of the QP subproblem at stage $k = 0, \dots, N-1$ reads

$$\begin{aligned}
H_k &= \nabla^2 \mathcal{L}(x, u, \pi, \lambda) \\
&= \begin{bmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{bmatrix} + \sum_{j=1}^{n_x} (\pi_k)_j \nabla^2 [f_k]_j(x_k, u_k) + \sum_{j=1}^{n_h} (\lambda_k)_j \nabla^2 [h_k]_j(x_k, u_k),
\end{aligned} \tag{4.2}$$

where n_h is the number of nonlinear path constraints, and π_k and λ_k are the Lagrange multipliers associated with the dynamics and nonlinear constraints, respectively.

The definiteness of H_k is thus determined by two parts, a constant cost block $\left[\begin{smallmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{smallmatrix}\right]$ and an iterate-dependent cost from the two curvature terms. In the

latter, each component Hessian $\nabla^2[f_k]_j$ and $\nabla^2[h_k]_j$ is symmetric but carries no definiteness guarantee, and the associated multipliers $(\pi_k)_j$ and $(\lambda_k)_j$ may take arbitrary sign and magnitude during the SQP iterations.

The full Hessian blocks H_k are thus general symmetric and might be indefinite matrices, in particular when the multipliers are large. Consequently, the exact-Hessian QP subproblem is in general nonconvex.

As for H_N, the dynamics add no curvature in x_N, but the nonlinear terminal constraints do, so

$$H_N = \mathcal{Q}_N + \sum_{j=1}^{n_{h_N}} (\lambda_N)_j \nabla^2[h_N]_j(x_N),$$

where n_{h_N} is the number of nonlinear terminal constraints and λ_N are their multipliers. Like H_k, H_N is thus iterate-dependent and possibly indefinite.

Collecting the stage Hessians H_k and Jacobians and linearizing the constraints of (4.1) at the current SQP iterate yields a structured QP. The subproblem takes the form of

$$\begin{aligned}
\underset{\Delta x,\, \Delta u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left(\frac{1}{2} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix}^\top \begin{bmatrix} Q_k & S_k^\top \\ S_k & R_k \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} + \begin{bmatrix} q_k \\ r_k \end{bmatrix}^\top \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} \right) \\
& \quad + \tfrac{1}{2}\, \Delta x_N^\top Q_N\, \Delta x_N + q_N^\top \Delta x_N \\
\text{subject to} \quad & \Delta x_0 = \Delta \bar{x}_0, \\
& \Delta x_{k+1} = A_k \Delta x_k + B_k \Delta u_k + c_k, \qquad k = 0, \dots, N-1, \\
& \begin{bmatrix} \underline{u}_k \\ \underline{x}_k \\ \underline{d}_k \end{bmatrix} \leq \begin{bmatrix} J_k^{b,u} & 0 \\ 0 & J_k^{b,x} \\ D_k & C_k \end{bmatrix} \begin{bmatrix} \Delta u_k \\ \Delta x_k \end{bmatrix} \leq \begin{bmatrix} \overline{u}_k \\ \overline{x}_k \\ \overline{d}_k \end{bmatrix}, \quad k = 0, \dots, N,
\end{aligned} \tag{4.3}$$

where, at each stage k, $\Delta x_k \in \mathbb{R}^{n_x}$ and $\Delta u_k \in \mathbb{R}^{n_u}$ denote the state and control deviation variables at stage k. Note that Δx_k and Δu_k here denote the SQP step, which the QP solver treats as its decision variables.

All remaining quantities are obtained by linearizing the OCP at the current SQP iterate (x_k, u_k) with multipliers (π_k, λ_k).

In the objective, Q_k, R_k and S_k are the state, control and coupling blocks of the exact Lagrangian Hessian H_k, and the stage-cost gradients are $q_k = \mathcal{Q}_k(x_k -$

$x_k^{\text{ref}}) + \mathcal{S}_k^\top (u_k - u_k^{\text{ref}})$ and $r_k = \mathcal{R}_k(u_k - u_k^{\text{ref}}) + \mathcal{S}_k(x_k - x_k^{\text{ref}})$, with Q_N and q_N as the terminal Hessian and gradient.

In the equality constraints, $A_k = \nabla_{x_k} f_k$ and $B_k = \nabla_{u_k} f_k$ are the Jacobians of the discretized dynamics, $c_k = f_k(x_k, u_k) - x_{k+1}$ is the dynamics residual, which vanishes at a dynamically feasible iterate, and $\Delta\bar{x}_0 = \bar{x}_0 - x_0$ is the residual of the initial state constraint $x_0 = \bar{x}_0$.

In the inequality constraints, $J_k^{b,u}$ and $J_k^{b,x}$ are selection matrices that pick out the bounded controls and states, with bounds $\underline{u}_k, \overline{u}_k$ and $\underline{x}_k, \overline{x}_k$ taken relative to the current iterate, like $\underline{d}_k, \overline{d}_k$ below. The matrices $C_k = \begin{bmatrix} \mathcal{C}_k \\ \nabla_{x_k} h_k \end{bmatrix}$ and $D_k = \begin{bmatrix} \mathcal{D}_k \\ \nabla_{u_k} h_k \end{bmatrix}$ stack the Jacobians of the affine and nonlinear path constraints, with the bounds

$$\underline{d}_k = \begin{bmatrix} \underline{g} - \mathcal{C}_k x_k - \mathcal{D}_k u_k \\ \underline{h} - h_k(x_k, u_k) \end{bmatrix}, \qquad \overline{d}_k = \begin{bmatrix} \bar{g} - \mathcal{C}_k x_k - \mathcal{D}_k u_k \\ \bar{h} - h_k(x_k, u_k) \end{bmatrix}.$$

At the terminal stage $k = N$, no control is present, so $J_N^{b,u}$ and D_N are empty, and C_N, $\underline{d}_N$ and $\overline{d}_N$ are formed in the same way from $h_N(x_N)$ with the bounds $\underline{h}_N, \bar{h}_N$.

4.1.2 Benchmark OCPs

In this subsection, we introduce the eight OCPs used to construct the benchmark. Each OCP is based on a dynamic system provided in [28] and formulated as a setpoint regulation problem with problem-specific modifications, which can be uniformly described with (4.1).

The eight OCPs are named and characterized as follows:

- **Pendulum:** Swing-up of an inverted pendulum on a cart under an input bound and a mixed state–input constraint.
- **Acrobot:** Adapted from [29]. An underactuated two-link pendulum moved between two prescribed configurations.
- **Dubins car:** Adapted from [30]. Parallel parking of a kinematic unicycle in a narrow corridor.
- **Furuta:** Adapted from [31]. Swing-up of a rotary inverted pendulum under tight torque limits.

- **Flying robot:** Adapted from [32]. A planar free-flying robot steered by four nonnegative thrusters.
- **Robot arm:** Adapted from [32]. A three-degree-of-freedom arm moved to a new set-point, with nonlinear input coupling.
- **Crane:** Adapted from [32]. An overhead crane moving a load between two rest positions.
- **Moonlander:** Adapted from [33]. Soft landing of a planar lunar lander with two thrusters.

Table 4.1: Summary of the evaluated OCPs in common control problem scales.

Problem	n_x	n_u	N
Pendulum	4	1	25
Acrobot	4	1	15
Dubins car	3	2	25
Furuta	4	1	20
Flying robot	6	4	50
Robot arm	6	3	30
Crane	6	2	40
Moonlander	6	2	30

Besides the different physical systems, these OCPs also cover different state and control dimensions, as shown in Table 4.1. The state dimension n_x ranges from 3 to 6, the control dimension n_u from 1 to 4, and the horizon length N from 15 to 50.

4.1.3 Collection pipeline

Based on the eight OCPs, we can construct a benchmark of indefinite QPs arising from SQP-based NMPC. Specifically, we establish a collection pipeline that extracts the QP subproblems directly from intermediate SQP iterations. The collection pipeline consists of two steps: *generate* and *collect*, and can be summarized as follows:

1. **Generate:** In this step, we generate iterates by solving the OCP with SQP and obtain iterates from early to late SQP iterations. To keep the SQP iterations from failing on indefinite exact-Hessian QPs, we use SQP-level regularization or the Gauss–Newton Hessian approximation in `acados`.

2. **Collect:** In this step, we utilize the previously generated iterates as initial guesses for linearization within the solver. We can selectively collect indefinite QPs that exhibit an indefinite reduced Hessian.

In the collect step, HPIPM uses the classical Riccati recursion (`ric_alg = 0`) [34], which requires only the reduced Hessian to be positive definite. A failure of the recursion thus indicates an indefinite reduced Hessian, which `acados` reports as status `4` (infeasible QP), and only the QPs with this status are collected.

For each problem, we keep only the QPs that the reference solver IPOPT [17] solves successfully, and limit the number of near-identical QPs, identified by comparing their solutions, so that the number of instances per problem stays within a comparable range and the collected instances remain meaningful. In the end, we obtain a total of 170 indefinite QPs from the eight OCPs. The per-problem counts are reported in Table 4.2 and range from 18 to 27. These QPs are stored in JSON format and are available at `https://github.com/Pandatheon/Primal-dual_IPM.git`.

Table 4.2: Number of indefinite QPs collected from each OCP (170 in total).

Problem	**Number**
Pendulum	18
Acrobot	19
Dubins car	22
Furuta	19
Flying robot	20
Robot arm	24
Crane	27
Moonlander	21

4.2 Benchmark analysis

In this section, we analyze the collected instances regarding the definiteness in the benchmark. Section 4.2.1 describes the magnitude and coverage of the indefinite reduced Hessians in the collected QPs. Section 4.2.2 further illustrates the source of indefiniteness in the collected QPs by analyzing the eigenvalues of the reduced Hessian blocks at each stage.

4.2.1 Distribution of indefiniteness

We first characterize the collected QPs along two axes: the *depth* of indefiniteness, measured by the magnitude of the most negative eigenvalue $|\nu_{\min}|$ of the reduced Hessian, and the *width*, measured by the fraction of reduced Hessian blocks that are indefinite within a QP.

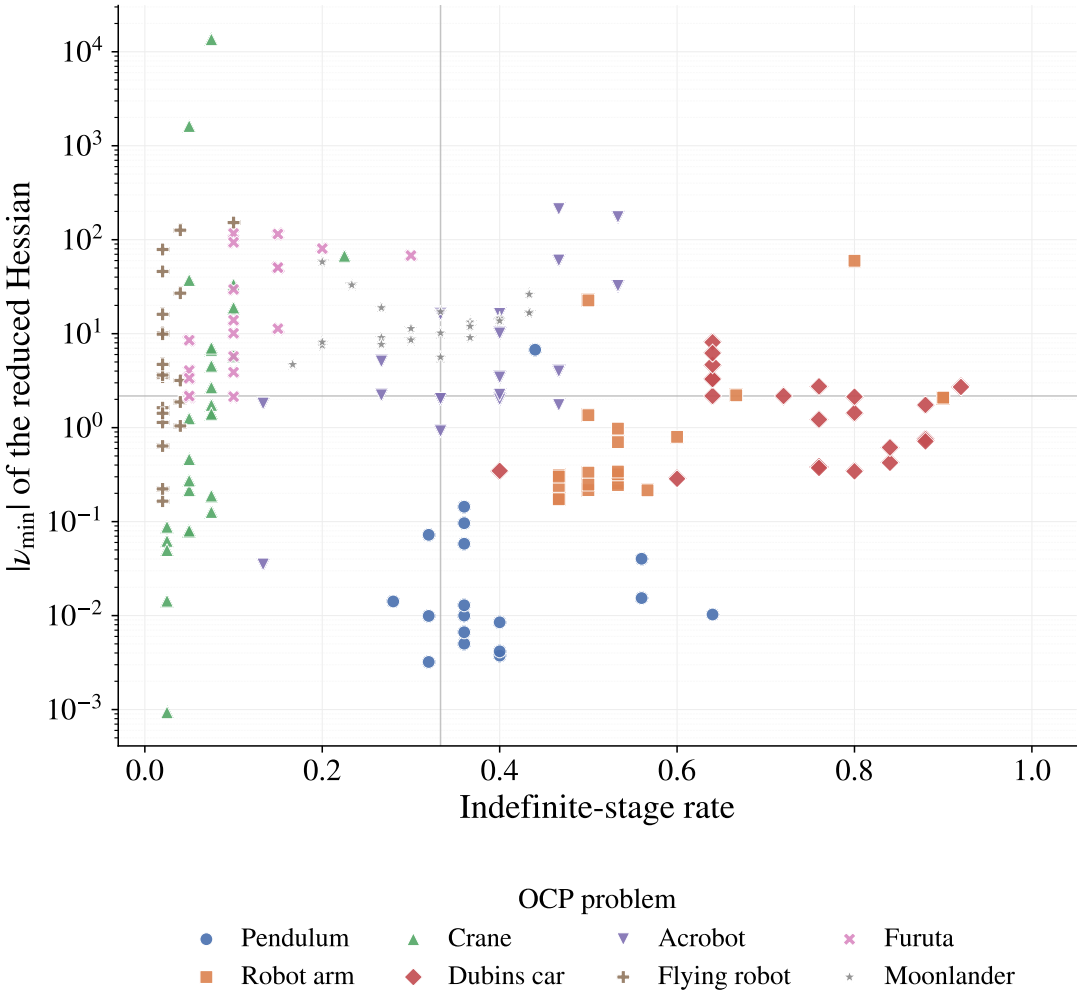


Figure 4.1: Depth ($|\nu_{\min}|$) versus width (indefinite-stage rate) of the collected QPs, colored by OCP. Grey lines mark the medians.

As seen from Figure 4.1, the depth of the collected QPs ranges from near zero to very large values, and the width from a few stages to almost the entire horizon. But wide and deep indefiniteness seldom occur together. Problems that are indefinite in only a small fraction of their stages, such as *Crane*, *Flying robot* and *Furuta*, can still reach deep negative curvature. Conversely, *Dubins car* and *Robot*

arm show only moderate depth with wide indefinite-stage rates. *Moonlander* and *Acrobot* lie in between, with intermediate indefinite-stage rates. *Moonlander* stays within a narrow depth band, whereas *Acrobot* spreads over a wide one. *Pendulum* contributes QPs with very shallow negative curvature at intermediate rates. The eight OCPs thus cover narrow-but-deep, intermediate and wide-but-moderate indefiniteness.

4.2.2 Source of indefiniteness

Recall that the reduced Hessian block, $\tilde{M}_k = \tilde{R}_k + B_k^\top P_{k+1} B_k$, at stage k, is assembled from two parts, the control cost block $\tilde{R}_k$ and the cost-to-go block P_{k+1}.

To investigate where the indefiniteness originates, we classify every indefinite stage by which of R_k and P_{k+1} has a negative eigenvalue: *R only* ($\tilde{R}_k \not\succeq 0$, $P_{k+1} \succeq 0$), *P only* ($P_{k+1} \not\succeq 0$, $R_k \succeq 0$), and *both* ($\tilde{R}_k, P_{k+1} \not\succeq 0$). Since $\tilde{R}_k, P_{k+1} \succeq 0$ would imply $M_k \succeq 0$, these three cases cover all indefinite stages.

Figure 4.2 shows this classification for each OCP. In every OCP, most indefinite stages involve P, either alone or together with R, and six of the eight OCPs are dominated by a single category, with only *Acrobot* and *Crane* mixed.

The *both* case dominates *Moonlander*, *Robot arm*, *Dubins car* and *Furuta*. The *P only* case dominates *Flying robot* and *Pendulum*, and is also the largest category in the two mixed problems, *Crane* and *Acrobot*, ahead of *both*. The *R only* case appears in three problems and is substantial only for *Dubins car*, while marginal in *Robot arm* and *Crane*.

Among all, *Flying robot* is the clearest case: all of its indefinite stages are *P only*, with $\tilde{R}_k \succ 0$ throughout. Its negative curvature thus enters the reduced Hessian only through the cost-to-go block, at isolated stages.

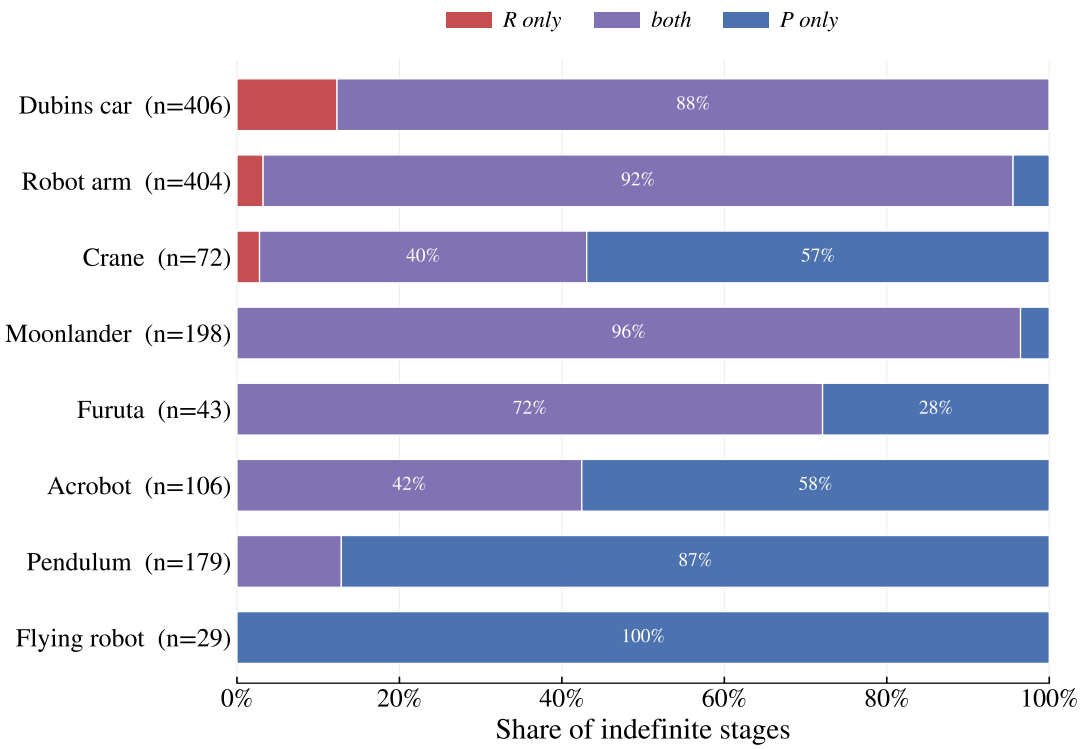


Figure 4.2: Classification of the indefinite stages of each OCP into *R only*, *both* and *P only* (n is the number of collected indefinite stages).

Chapter 5

Numerical experiments

In this chapter, we evaluate the proposed regularization methods numerically and discuss the results. We begin in Section 5.1 with the two metrics used throughout the comparison, the iteration profile and the flop unit estimation, which measure convergence behavior and factorization cost respectively. Section 5.2 then describes the prototype solver on which all experiments are run and stresses that it reproduces the iteration behavior of HPIPM on convex QPs, so that any difference observed later can be attributed to the regularization rather than to the solver itself. Section 5.3 fixes the experiment setup, naming and classifying the regularization methods under comparison and stating the parameters shared by all of them. Section 5.4 presents the numerical comparison, first among the Riccati-based methods themselves and then against full Hessian regularization and inertia correction, followed by ablations of the barrier update and the line search. Section 5.5 concludes the chapter by analyzing the failures and the Mehrotra ablation.

5.1 Experiment metrics

In this section, we introduce the metrics used to evaluate the performance of different regularization strategies in our experiments. Since we care about both the convergence behavior and the factorization cost caused by different regularization methods, we use two complementary metrics: iteration profile and flop unit estimation. The former, iteration profile, is used to evaluate how many iterations

each method takes to converge on a set of test instances, while the latter, flop unit estimation, is approximated by the modified Cholesky factorizations of the reduced Hessian and is used to evaluate the factorization cost of each method.

5.1.1 Iteration profile

The *iteration profile* shows the number of test instances that the method solves within a certain iteration budget. An instance that the method fails to solve within the max iteration budget $i_{\max}$ counts as unsolved.

We plot the iteration profile as a curve, with the iteration budget from 1 to $i_{\max}$ on the horizontal axis and the number of solved instances on the vertical axis. Different from the common performance profile [35], which uses the iteration ratio to the best method as the horizontal axis, the iteration profile of a method does not depend on the other methods, by which we emphasize the ability of each method itself to solve the instances.

Plotting the profiles of all methods in one figure allows a direct comparison: at any budget, the higher curve belongs to the method that has solved more instances. And a curve that rises steeply at small budgets indicates an *efficient* method, and its final value at $i_{\max}$, the number of solved instances, measures its *robustness*.

5.1.2 Flop unit estimation

The regularization strategies differ in the factorization they perform per iteration. Some use an eigenvalue decomposition, some a modified Cholesky factorization, and some repeat the Cholesky factorization with an increasing diagonal shift. Iteration counts alone therefore do not reflect the factorization cost, so we express this cost in a common proxy unit.

We take one Cholesky factorization of the reduced Hessian block as the proxy unit. For factorizing an $n_u \times n_u$ matrix, the cost is approximately $\frac{1}{3}n_u^3$ flops, hence

$$\text{unit} \;\approx\; \tfrac{1}{3}\, n_u^3. \tag{5.1}$$

The modified factorization only adds diagonal updates of $\mathcal{O}(n_u^2)$ cost, so to leading order it also costs one unit.

The other factorizations and decompositions are expressed relative to this unit. A symmetric eigenvalue decomposition of the same block costs about $9n_u^3$ flops [36], i.e.

$$\frac{9n_u^3}{\frac{1}{3}n_u^3} \;=\; 27 \text{ units.} \tag{5.2}$$

Regarding the cost-to-go matrix with dimension $n_x \times n_x$ and a full Hessian block with dimension $(n_x + n_u) \times (n_x + n_u)$, the ratios are considered as

$$\omega_P \;=\; \left(\frac{n_x}{n_u}\right)^3, \quad \omega_H \;=\; \left(\frac{n_x + n_u}{n_u}\right)^3. \tag{5.3}$$

Thus, a modified Cholesky factorization of the cost-to-go matrix costs ω_P units and an eigenvalue decomposition $27\omega_P$ units.

Likewise, the two factorizations of a full Hessian block cost ω_H and $27\omega_H$ units. Since the full Hessian regularization modifies all N blocks $\tilde{H}_k$ at once, each regularization costs $N\omega_H$ or $27N\omega_H$ units.

The factorization cost of a run is thus the sum of two parts, the Cholesky factorizations of the Riccati recursion itself and the additional factorizations introduced by the regularization. The latter includes every rerun of the Riccati recursion, whose factorizations are counted again, and every failed Cholesky factorization, which counts as one proxy unit, although it may stop early.

5.2 Prototype solver implementation

To ensure a fair comparison, we implemented all regularization methods in one prototype solver. The Riccati-based primal–dual IPM prototype solver is implemented in Python, using HPIPM as a reference. The prototype was designed to reproduce its iteration behavior on convex QPs and is available at `https://github.com/Pandatheon/Primal-dual_IPM.git`.

One implementation detail deserves a special mention, as it deviates from the plain FTB rule stated in Section 2.4. Rather than being a fixed fraction η of the maximal step, the primal and dual step sizes α_{prim} and α_{dual} are obtained from their respective maximal steps through the heuristic

$$\alpha = \begin{cases} \alpha_{\max}\left[(1-\alpha_{\max})\,\kappa_\eta + \alpha_{\max}\hat{\eta}\right], & \alpha_{\max} < 1, \\ \alpha_{\max}, & \alpha_{\max} = 1, \end{cases} \tag{5.4}$$

where $\alpha_{\max} \in (0, 1]$ denotes the maximal step length from (2.22) or (2.23), $\kappa_\eta = 0.99$ and $\hat{\eta} = 1 - 10^{-7}$ as in HPIPM.

When $\alpha_{\max} < 1$, the bracketed factor changes with $\alpha_{\max}$, from a strongly blocked step to a nearly full one, so the closer the maximal step is to a full step, the less it is shortened. When $\alpha_{\max} = 1$, an unblocked full step can be taken without scaling by $\hat{\eta}$ so that the local quadratic convergence rate of the IPM can be preserved.

We further align the prototype solver with HPIPM in terms of the mode settings by defining a set of HPIPM settings, referred to as the following presetting, whose values are listed in Table 5.1.

Table 5.1: Detailed settings for the defined presetting.

Parameter	**Value**	**Parameter**	**Value**
mu0	1e1	lam_min	1e-12
alpha_min	0.0	t_min	1e-12
res_g_max	1e-6	tau_min	1e-12
res_b_max	1e-8	lam0_min	1e-12
res_d_max	1e-8	t0_min	1e-12
res_m_max	1e-6	warm_start	0
dual_gap_max	1e15	abs_form	0
iter_max	30	comp_dual_sol_eq	1
stat_max	30	comp_res_exit	1
pred_corr	1	comp_res_pred	1
cond_pred_corr	1	split_step	1
itref_pred_max	0	m_safe	0.5
itref_corr_max	0	var_init_scheme	0
reg_prim	0.0	t_lam_min	2
square_root_alg	0	t0_init	2
lq_fact	0	update_fact_exit	1

As one can notice, the presetting largely coincides with the `SPEED` mode of

HPIPM, differing in three respects. First, the square-root algorithm is disabled (`square_root_alg`= 0) so that the Riccati recursion requires no positive definite full Hessian to carry on. Second, the primal and dual step sizes are computed separately (`split_step`= 1) instead of sharing a single step length. Third, the minimal step size is disabled by setting `alpha_min`= 0. In both modes no iterative refinement is applied (`itref_pred_max`=`itref_corr_max`= 0) and the conditional Mehrotra's predictor–corrector is retained (`pred_corr`=`cond_pred_corr`= 1).

To verify that the two implementations do agree, we collected 40 convex QP subproblems, five from each of the 8 OCPs in the benchmark, and compared the iteration behavior of the prototype solver against that of HPIPM with the presetting. The resulting iteration profiles are shown in Figure 5.1.

Both solvers solve all 40 instances in the exact same number of iterations for each instance, resulting in iteration profiles that coincide. On convex QPs, the prototype solver therefore reproduces the iteration behavior of HPIPM, which verifies the prototype as a reliable baseline and allows us to attribute the differences observed in the experiments to the regularization rather than to the solver.

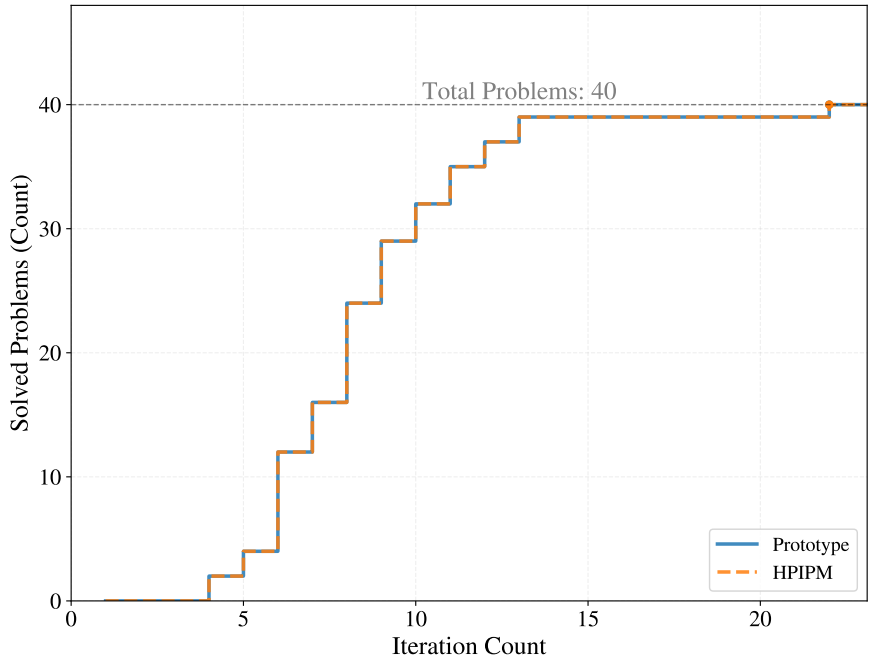


Figure 5.1: Iteration profiles of the prototype solver and HPIPM on 40 convex QPs. The two curves coincide.

5.3 Experiment setup

In this section, the settings for the experiments are introduced. First, based on the previously discussed methods, we name and classify the regularization methods considered in the experiment in Section 5.3.1, and then we introduce the general setup of the experiments in Section 5.3.2, including the convergence criterion and step size control strategy.

5.3.1 Evaluated regularizations

Before turning to the results, we fix the set of regularization methods entering the comparison, together with the naming convention used throughout this chapter.

Every method is named in the form TARGET_STRATEGY. The TARGET indicates the matrix or matrices from which the correction is derived and to which it is applied, which gives the three categories of *full Hessian regularization*, *Riccati-based regularization*, and *inertia correction*. The STRATEGY is one of the numerical strategies of Section 3.1.1, which are largely independent of the matrix they act on and can therefore be paired with each target.

Full Hessian regularization. The first category follows Section 3.1.3 and takes the full Hessian as the target matrix. The regularization is triggered by the failure of the Cholesky factorization in the Riccati recursion, and is applied stage-wise to each full Hessian block. The Riccati recursion is then re-attempted with the regularized full Hessian. Applying the three strategies of Section 3.1.1 to each block $\tilde{H}_k$ gives H_PROJECT, H_MIRROR and H_GMW.

Riccati-based regularization. The second category is the one proposed in Section 3.2 and acts on the reduced Hessian $\tilde{M}$. The algorithms are given in Algorithms 5 and 6, which differ in which matrices of the backward pass the adjustment is derived from and applied to, giving two targets within this category:

- M: only the reduced Hessian block $\tilde{M}_k$ is corrected, as in Algorithm 5. The correction is purely local and serves as a minimal safeguard for the

factorization.

- PM: the cost-to-go matrix P_{k+1} entering the stage is corrected as well, as in Algorithm 6 introduced in this work, so that the analyzed propagation of small curvature is taken into account.

Pairing each target with the three strategies gives six methods, which correspond to the two insights of Section 3.3. M_PROJECT applies neither of the two and serves as the reference, M_MIRROR and M_GMW apply only the first insight, i.e. non-minimum perturbation, PM_PROJECT applies only the second, which considers the propagation with minimum perturbation, and PM_MIRROR and PM_GMW apply both.

Inertia correction. The third category is the Levenberg–Marquardt-style inertia correction of Section 3.1.1, which we denote by INERTIA_LM.

This category is distinguished from the two preceding categories, as it has no distinct target matrix in the above sense. In this category, a scalar multiple of the identity $\delta_w I$ is added to each full Hessian block $\tilde{H}_k$, which is equivalent to adding it to the full Hessian, i.e. $\tilde{H}_i + \delta_w I$, and the Riccati recursion is repeated with increasing δ_w until it succeeds. The trial sequence follows the inertia correction of IPOPT [17] and is given in Algorithm 7, with the IPOPT defaults listed in Table 5.2. Only the primal regularization from IPOPT is retained, as dual regularization is not the primary focus of this work.

Table 5.2: Inertia correction parameters.

Symbol	Value	Description
δ_w^0	10^{-4}	Initial Hessian perturbation parameter
$\delta_w^{\min}$	10^{-16}	Minimum allowable regularization parameter
$\delta_w^{\max}$	10^{16}	Maximum allowable regularization threshold
κ_w^-	$1/3$	Factor to decrease perturbation upon success
κ_w^+	8	Factor to increase perturbation after inertia failure
$\bar{\kappa}_w^+$	10^2	Accelerated scaling factor for severe indefiniteness

Taken together, the three categories span ten methods, which all run on the same solver, same benchmark and with the same settings.

Algorithm 7 INERTIA_LM, following IPOPT [17]

Input: $\tilde{H}_i$, δ_w^{last}

Riccati recursion with $\tilde{H}_i$
if succeeds **then**
    $\delta_w^{\text{last}} \leftarrow 0$
    return
end if
if $\delta_w^{\text{last}} = 0$ **then**
    $\delta_w \leftarrow \delta_w^0$
    $\kappa \leftarrow \bar{\kappa}_w^+$
else
    $\delta_w \leftarrow \max\{\delta_w^{\min}, \kappa_w^- \delta_w^{\text{last}}\}$
    $\kappa \leftarrow \kappa_w^+$
end if
while $\delta_w \leq \delta_w^{\max}$ **do**
    Riccati recursion with $\tilde{H}_i + \delta_w I$
    if succeeds **then**
        $\delta_w^{\text{last}} \leftarrow \delta_w$
        return
    end if
    $\delta_w \leftarrow \kappa\, \delta_w$
end while
return failure

5.3.2 General setup

The parameter settings regarding experiments are collected in Table 5.3. The implemented IPM is summarized in Algorithm 8, which uses Mehrotra's probing for updating barrier parameters, and the two-type line search to control step size after the FTB rule.

Table 5.3: Settings shared by all experiments.

Symbol	Value	Role
$\varepsilon_{\text{stat}}$	10^{-6}	Tolerance on the stationarity residual
ε_{eq}	10^{-8}	Tolerance on the equality residual
$\varepsilon_{\text{ineq}}$	10^{-8}	Tolerance on the inequality residual
$\varepsilon_{\text{comp}}$	10^{-6}	Tolerance on the complementarity residual
$i_{\max}$	120	Iteration budget
$n_{\mathcal{P}}$	170	Nonconvex QP instances in the test set
ϵ	10^{-4}	Eigenvalue and pivot floor
β	0.4	Backtracking factor, $\alpha_{\text{prim}} \leftarrow \beta\alpha_{\text{prim}}$
$n_{\max}$	10	Maximum backtracking iterations
κ_s	10^{-1}	Switching constant, $\alpha_{\text{prim}}(-\rho) > \kappa_s\,\theta^2$
γ	10^{-4}	Armijo constant, $\phi^+ \leq \phi_i + \gamma\,\alpha_{\text{prim}}\rho$

Algorithm 8 Primal–dual IPM used in the experiments

Input: QP with OCP structure, initial iterate, iteration budget $i_{\max}$
for $i = 0$ **to** $i_{\max} - 1$ **do**
  compute $r_i^{\text{stat}}, r_i^{\text{eq}}, r_i^{\text{ineq}}, r_i^{\text{comp}}$
  if all residuals are below $\varepsilon_{\text{stat}}, \varepsilon_{\text{eq}}, \varepsilon_{\text{ineq}}, \varepsilon_{\text{comp}}$ **then**
    return current iterate
  end if
  factorize LHS with regularizations ▷ Proxy units recorded
  Riccati recursion with $r_i^{\text{comp,aff}}$ for the affine direction
  compute μ_{aff} and $\sigma_i \leftarrow (\mu_{\text{aff}}/\mu_i)^3$ ▷ Mehrotra's probing
  Riccati recursion with $r_i^{\text{comp,pro}}$ for the search direction
  compute $\alpha_i^{\text{prim}}, \alpha_i^{\text{dual}}$ by the FTB
  line search to reduce α_i^{prim} ▷ two-type line search
  update primal and dual variables
end for
return Max iteration reached

5.4 Numerical comparison

In this section, we report the numerical results of the ten regularization methods on the nonconvex QPs of the benchmark. The comparison proceeds in three stages, each narrowing the field. We first compare the six Riccati-based methods against one another, in order to identify the strongest representatives of the category. These are then set against the full Hessian regularizations, and finally against inertia correction. Two ablations follow, which replace Mehrotra's probing with the conditional predictor–corrector in Section 5.4.4 and disable the line search in Section 5.4.5.

5.4.1 Comparison among Riccati-based regularizations

We first compare the proposed Riccati-based regularization methods, which are the main focus of this work. As a baseline we also include the unregularized recursion (NO_REGULARIZE).

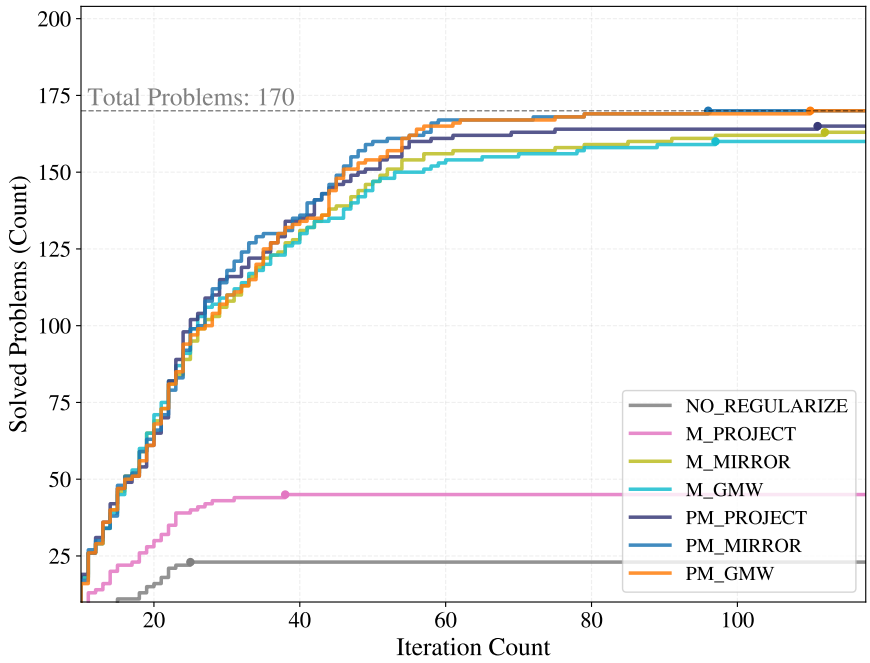


Figure 5.2: Iteration profile of the six Riccati-based methods and the unregularized baseline over the 170-instance test set.

In the iteration profile of Figure 5.2, two trends stand out. First, each PM

dominates its counterpart M once the budget exceeds about 40 iterations. Second, among the regularization strategies, MIRROR and GMW are clearly more effective than PROJECT. These trends validate the insights mentioned in Section 3.3. Among all the methods, PM_MIRROR and PM_GMW top the profile.

Besides, it is worth noting that the unregularized recursion can also solve a non-negligible fraction of the test set, around 25 of the 170 instances, which can be explained by the curvature contribution from the barrier term that makes the reduced Hessian positive definite in some instances.

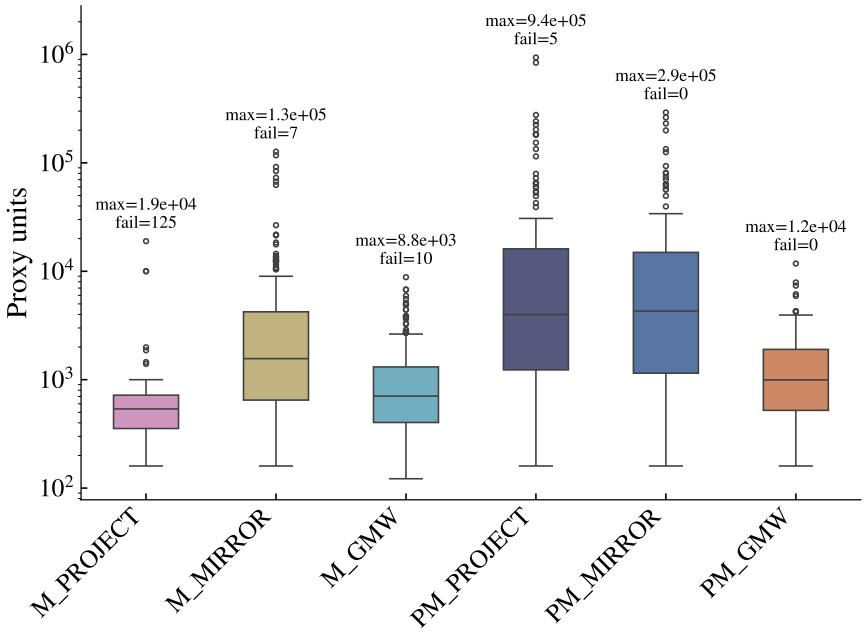


Figure 5.3: Factorization cost per instance of the six Riccati-based methods, unsolved instances due to maximal iteration included. Annotations report the maximum observed cost and the number of unsolved instances.

Figure 5.3 shows the distribution of the factorization cost per instance in proxy units for each method. Instances that reach the iteration cap $i_{\max}$ are included with the cost of all $i_{\max}$ iterations, so the distribution reflects the worst-case cost.

In Figure 5.3, all methods share a similar lowest cost, since some instances are solved by the unregularized recursion. The methods based on eigenvalue modification cost more than those based on modified Cholesky factorization as expected, and the PM methods, which also act on P_{k+1}, cost more than their M counterparts.

Taken together, the two figures first reveal a robustness ordering, {PM_MIRROR, PM_GMW} > {M_MIRROR, M_GMW, PM_PROJECT} > M_PROJECT. Since PM_MIRROR and PM_GMW solve all instances, we therefore focus on the two Riccati-based regularizations in the following comparisons with other regularization categories.

5.4.2 Comparison with full Hessian regularizations

Having identified PM_MIRROR and PM_GMW as the strongest Riccati-based methods, we now compare them against the full Hessian regularizations.

The iteration profile in Figure 5.4 shows that the two Riccati-based methods sit well above every full Hessian variant across the whole iteration budget, which indicates better performance in efficiency and robustness of the two Riccati-based methods over full Hessian regularization.

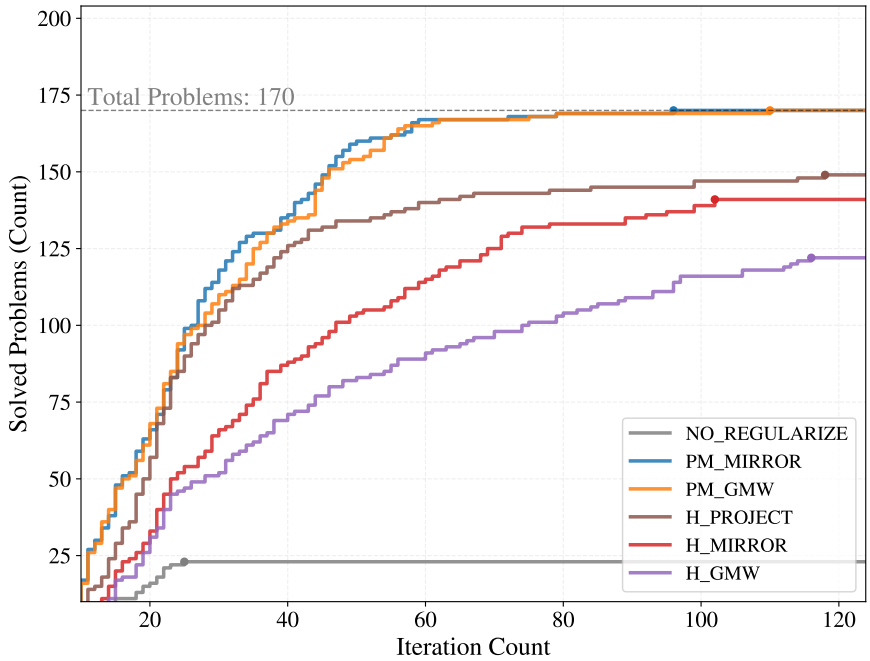


Figure 5.4: Iteration profile of the two strongest Riccati-based methods (PM_MIRROR, PM_GMW) against the full Hessian regularizations (H_PROJECT, H_MIRROR, H_GMW) and the unregularized baseline.

In Figure 5.5, the two PM methods also outperform them on the factorization cost. The worst-case costs of H_PROJECT and H_MIRROR lie far above those

of both Riccati-based methods, and even H_GMW, the cheapest and least robust full Hessian method, has a higher median cost than PM_GMW.

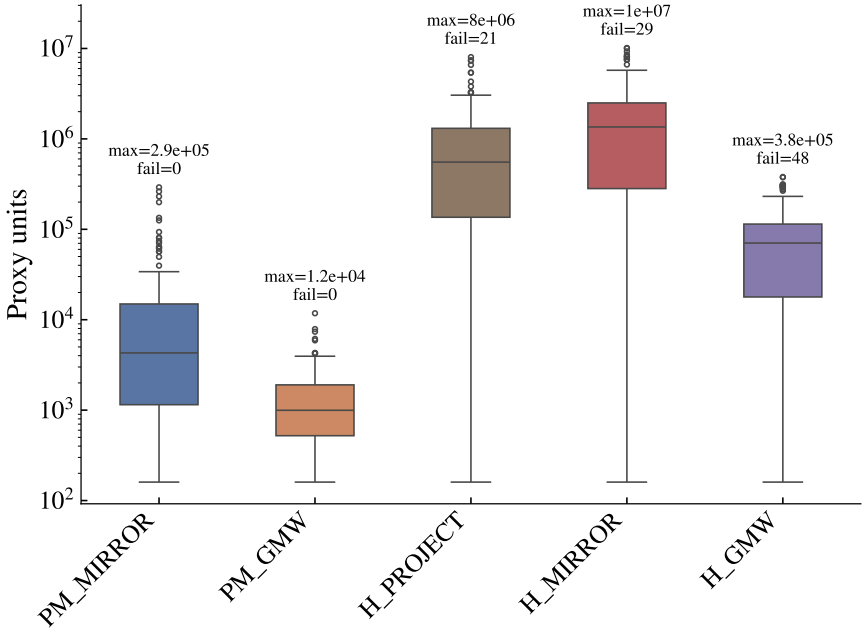


Figure 5.5: Factorization cost per instance of the two strongest Riccati-based methods against the full Hessian regularizations. Unsolved instances and annotations as in Figure 5.3.

The Riccati-based methods are thus generally more robust and cheaper than every full Hessian counterpart. This supports regularizing the reduced Hessian by exploiting the stage-wise structure of the problem rather than the full Hessian.

5.4.3 Comparison with inertia correction

Finally, we compare the two Riccati-based methods against inertia correction (INERTIA_LM).

In Figure 5.6, PM_MIRROR and PM_GMW still sit above INERTIA_LM, which leaves 9 unsolved at the iteration cap. This indicates better robustness and efficiency of the two Riccati-based methods over INERTIA_LM with the default parameters in Table 5.2.

Figure 5.7 draws the distinction among the three methods more finely. Though the factorization cost of reruns is accounted for, inertia correction still has the

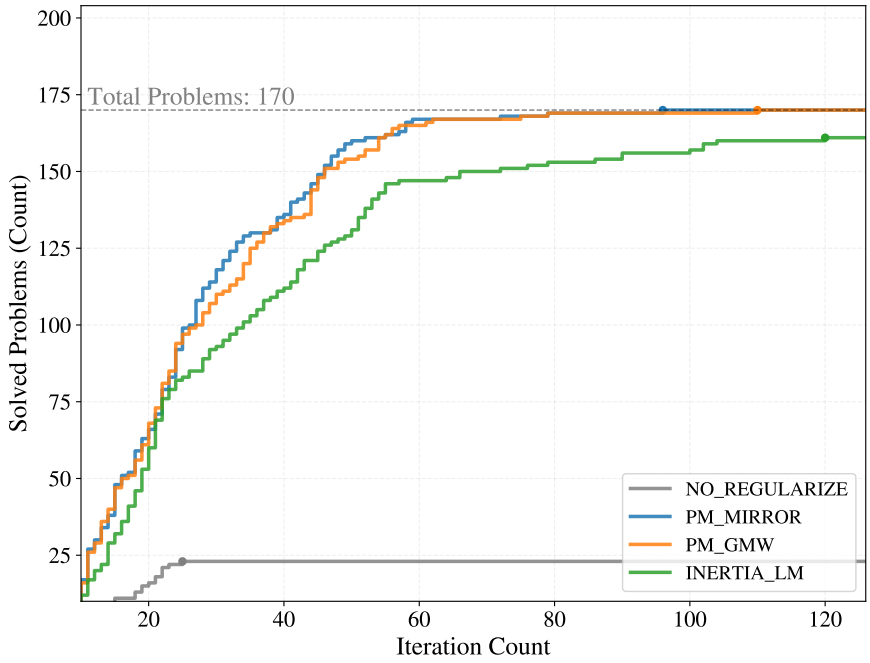


Figure 5.6: Iteration profile of PM_MIRROR and PM_GMW against inertia correction (INERTIA_LM) and the unregularized baseline.

lowest median cost of the three, slightly below PM_GMW. In the worst case, however, INERTIA_LM and PM_GMW are approximately on par.

From the evidence of this benchmark, inertia correction remains competitive in factorization cost, similar to PM_GMW.

5.4.4 Ablation on the conditional Mehrotra

All results in the previous sections used Mehrotra's probing. To justify this choice, we rerun PM_MIRROR, PM_GMW, INERTIA_LM and H_PROJECT, together with the unregularized baseline, using the conditional Mehrotra's predictor–corrector instead, with all other settings unchanged.

Comparing Figure 5.8 with the previous results shows that every method solves fewer instances with the corrector, but the loss is far from uniform. PM_MIRROR and PM_GMW drop drastically, whereas INERTIA_LM and H_PROJECT only drop about 10 and 5 cases. The ranking of Section 5.4 is thereby reversed, and the two Riccati-based methods sit at the bottom.

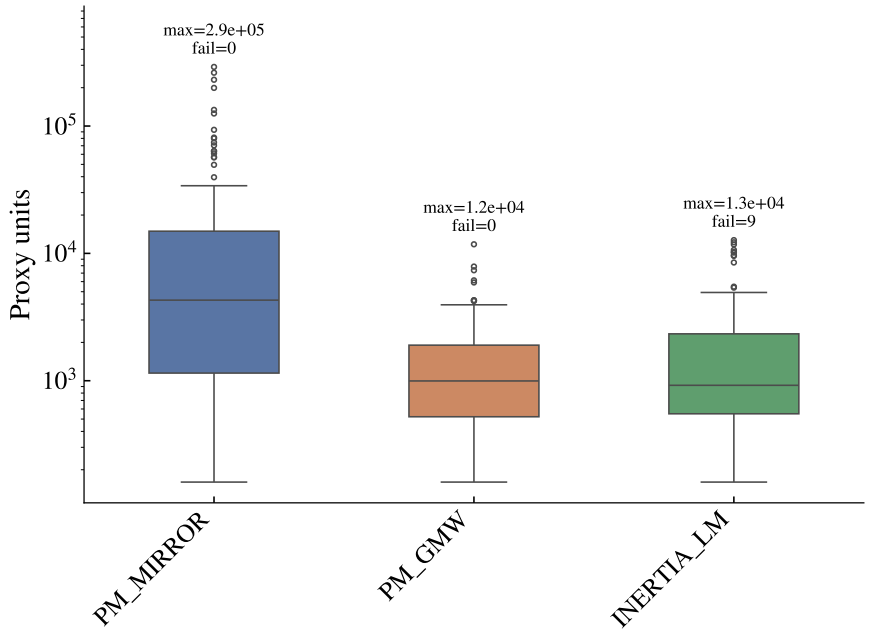


Figure 5.7: Factorization cost per instance of PM_MIRROR and PM_GMW against inertia correction. Unsolved instances and annotations as in Figure 5.3.

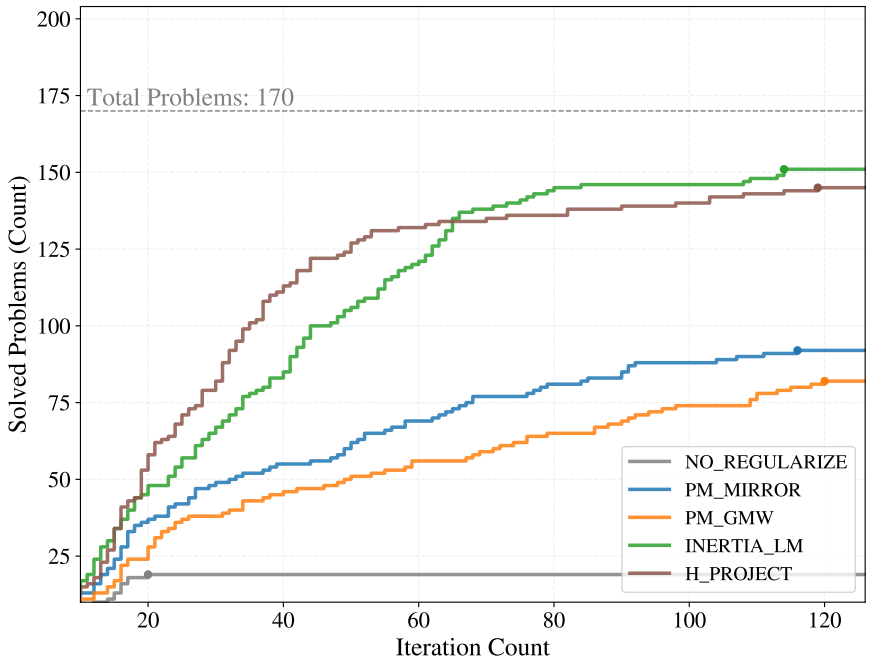


Figure 5.8: Iteration profile with the conditional predictor–corrector of PM_MIRROR, PM_GMW, INERTIA_LM, H_PROJECT and the unregularized baseline.

But generally speaking, this supports Mehrotra's probing as the default method for nonconvex QPs, as it improves the performance of methods generally. Section 5.5.3 discusses why PM_MIRROR loses far more instances than INERTIA_LM.

5.4.5 Ablation on the line search

The two-type line search of Section 2.4.2 is introduced for investigating the effect of line search on nonconvex QPs. To isolate its contribution, we disable it, so that the FTB step is taken directly, and rerun the same five methods as in the previous subsection with all other settings unchanged.

Table 5.4: Number of instances solved within 40 and 120 iterations without and with the line search.

	$i_{\max} = 40$		$i_{\max} = 120$	
Method	without	with	without	with
PM_MIRROR	137	136	170	170
PM_GMW	135	134	169	170
INERTIA_LM	112	112	161	161
H_PROJECT	127	126	149	149
NO_REGULARIZE	18	23	18	23

Table 5.4 shows that the line search matters much less than the barrier update. Only NO_REGULARIZE clearly benefits from the line search, while the regularized methods solve almost the same number of instances. The numbers solved within 40 iterations with each regularization are also almost identical, so the line search does not accelerate convergence either.

The ranking of Section 5.4 therefore does not rely on the line search. The robustness and efficiency of the regularization method stem from the method itself.

5.5 Discussion

This section first breaks down the failures in two ways. Section 5.5.1 splits them by termination status, separating numerical breakdowns of the regularized recur-

sion from exhausted iteration budgets. Section 5.5.2 splits them by the OCP each instance comes from, which shows what makes instances hard and why some methods perform poorly. Finally, Section 5.5.3 explains why the conditional Mehrotra's predictor–corrector affects PM_MIRROR and INERTIA_LM so differently.

5.5.1 Failure by status

A run that does not converge ends with one of three exit statuses, shown per method in Figure 5.9:

- *Unregularized*: the Cholesky factorization fails *again* after the regularization has been applied, usually caused by numerical indefiniteness.
- *Max iterations reached*: the residuals did not fall below their tolerances within the budget $i_{\max} = 120$.
- *NaN encountered*: an infinite or undefined value appeared in the residuals, so that the iterate itself has overflowed.

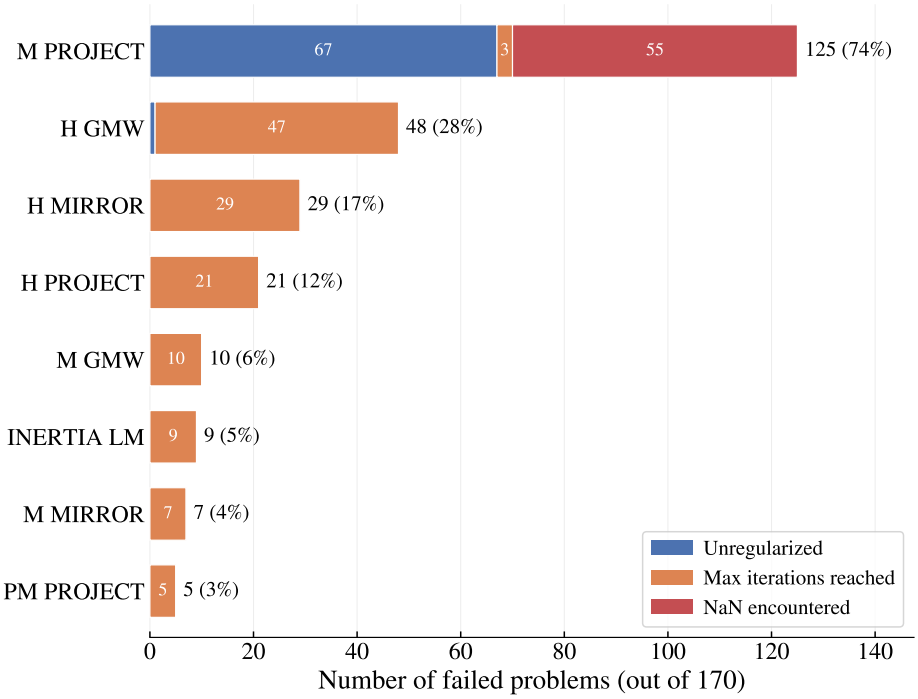


Figure 5.9: Unsolved instances per method on the 170 instances, excluding PM_MIRROR and PM_GMW, stacked by exit status and annotated with the total and its share of the test set.

In Figure 5.9, M_PROJECT is the only Riccati-based method with numerical breakdowns, with 122 of its 125 failures stemming from numerical issues. This confirms the analysis in Section 3.3, where ill-conditioning propagates into numerical indefiniteness or overflow.

Either insight introduced in Section 3.3 resolves these breakdowns. Abandoning the minimum perturbation rule (M_MIRROR, M_GMW) or adding the cost-to-go correction (PM_PROJECT) leaves only iteration-limit terminations. Combining both remedies, as in PM_MIRROR and PM_GMW, achieves the best performance in this benchmark.

Among the full Hessian methods, only H_GMW still breaks down, on a single instance, and all other failures reach the iteration limit. This breakdown comes only from the numerical indefiniteness during the Riccati recursion in the IPM. It remains a potential issue for full Hessian regularization, as rounding errors can still make the Cholesky factorization fail.

5.5.2 Failure by category

Figure 5.10 breaks down the same failures by the OCP each instance was collected from. Combined with Figure 4.1, M_MIRROR, M_GMW and PM_PROJECT fail mostly on the wide indefiniteness of *Dubins car* and *Robot arm*, whereas the full Hessian methods fail mostly on the narrow indefiniteness of *Flying robot* and *Crane*.

For the full Hessian regularizations and INERTIA_LM, *Flying robot* is the hardest OCP, and, as Figure 4.2 shows, its indefiniteness lies only in P_k and at few stages. H_PROJECT, H_MIRROR and H_GMW fail on 19, 20 and 20 of its 20 instances, and INERTIA_LM on 8, which are 8 of its 9 failures, while every Riccati-based method except M_PROJECT solves all of them.

Since the full Hessian regularizations and INERTIA_LM cannot directly act on P_k, they perturb the $\tilde{H}_k$ over the whole horizon. This slows down the convergence and thus exhausts the iteration budget. The Riccati-based regularization avoids this by treating an indefinite stage where it arises in the recursion and only affecting the preceding stages.

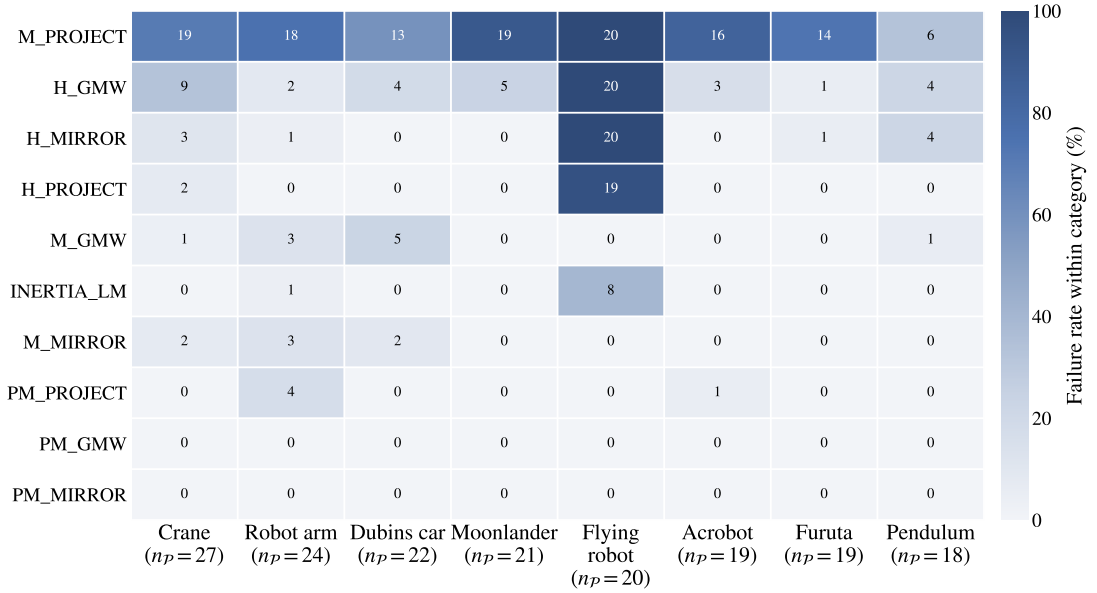


Figure 5.10: Unsolved instances per method and OCP with $n_\mathcal{P}$ as the number of instances collected from that OCP.

5.5.3 Analysis of the Mehrotra ablation

Based on the ablation in Section 5.4.4, this subsection explains why the ranking of Section 5.4 reverses under the conditional Mehrotra corrector, i.e. why PM_MIRROR loses far more instances than INERTIA_LM.

With the simplified notation of (2.9), the regularized system and its step bound read

$$(\tilde{M}_i + E_i)\,\Delta y_z = -r_i, \qquad \|\Delta y_z\| \leq \frac{\|r_i\|}{\nu_{\min}(\tilde{M}_i + E_i)}. \tag{5.5}$$

As one can notice, the regularization would shorten the step taken, and the methods differ in how small $\nu_{\min}(\tilde{M}_i + E_i)$ can become, and thus in how long the regularized step is.

INERTIA_LM increases a uniform shift δ_w until the factorization succeeds. If δ_w^* is the smallest shift for which it succeeds, then $\nu_{\min}(\tilde{M}_i + E_i) \geq \delta_w - \delta_w^*$. The accepted δ_w usually is larger than δ_w^*, and $\nu_{\min}$ stays of the order of δ_w. PM_MIRROR modifies only the negative eigenvalues of $\tilde{M}_k$ and P_k, which allows $\nu_{\min}(\tilde{M}_i + E_i)$ to stay as a small positive value. PM_MIRROR could thus reach a much longer step than INERTIA_LM at the same $\|r_i\|$.

As shown in Table 5.5, the median affine step of PM_MIRROR is longer than that of INERTIA_LM. This difference can be amplified in Mehrotra's predictor–corrector scheme, where the second-order term $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ grows quadratically with the affine step.

Table 5.5: Regularized step metrics of PM_MIRROR and INERTIA_LM.

	PM_MIRROR	INERTIA_LM
Corrector, on 170 *instances*		
Median affine step $\|\Delta y_z\|$	88.7	5.6
Percentage of rejected corrector steps	4.5 %	55.7 %
Probing, on 161 *instances solved by both*		
Total Iterations	4056	5261
Regularized iterations	1572	2808
Median displacement $\alpha_{\text{prim}} \|\Delta y_z\|$	1.61	0.38

In PM_MIRROR, the long affine step makes the second-order term dominate the RHS, so the corrector step is long as well. For both steps, the FTB rule then allows only a short step size, which keeps μ_{aff} and μ_{cor} close to μ. The condition $\mu_{\text{cor}} < 2\mu_{\text{aff}}$ thus rarely detects this. The corrector step dominated by the inexactness from an affine step is, therefore, taken, which cannot efficiently reduce the complementarity residual or the duality gap. In INERTIA_LM, by contrast, the second-order term is usually still comparable in size to the rest of the RHS, so the conditional Mehrotra's predictor–corrector scheme remains effective, and the condition can still reject the corrector steps as shown in Table 5.5. The scheme thus makes progress as intended, and INERTIA_LM loses far fewer instances than PM_MIRROR.

With probing, the affine step only affects the centering parameter, so the inexactness no longer enters the search direction. The longer step of PM_MIRROR, which is a drawback under the corrector, thus becomes an advantage, as measured by the displacement $\alpha_{\text{prim}} \|\Delta y_z\|$ in Table 5.5.

INERTIA_LM moves less per regularized iteration and generally needs more regularized iterations, while PM_MIRROR moves further per regularized iteration. This larger displacement could help the iterate to move towards the constraint boundaries, where the reduced Hessian could be positive definite with the curvature contribution, and recover the fast convergence towards solutions.

Chapter 6

Conclusion and outlook

This thesis investigated the regularizations of indefinite QP subproblems from exact-Hessian SQP for NMPC inside the interior-point solver, where the barrier curvature is already included and only the reduced Hessian has to be positive definite. Since the Riccati backward pass is a block $\mathrm{LDL}^\top$ factorization of the reduced Hessian, indefiniteness is detected and corrected stage-wise wherever the Cholesky factorization of $\tilde{M}_k$ fails, and the computational cost stays linear in the horizon length.

We showed that a minimum perturbation correction of $\tilde{M}_k$ adds a negative rank-one term to the cost-to-go matrix, which can make the preceding stages indefinite and ill-conditioned. The improved strategy therefore uses a non-minimal correction and also regularizes the cost-to-go matrix. We proved that the regularization becomes inactive close to the solution, so the local convergence rate of the IPM is retained.

With Mehrotra's probing, PM_MIRROR and PM_GMW solved all 170 instances, PM_GMW at a lower worst-case decomposition cost than PM_MIRROR, against 21 to 48 unsolved for full Hessian regularization and 9 for inertia correction. The full Hessian methods failed on almost every *Flying robot* instance, where only the cost-to-go matrix is indefinite, and only at a few stages.

Several questions remain open. First, the regularization magnitude, set by the eigenvalue floor ϵ, is fixed, and the remaining failures occur where large corrections are applied at many stages. Adapting the magnitude, e.g. to the conditioning of the corrected blocks or to the barrier parameter, could improve robustness. Second,

in time-optimal and economic MPC, the cost provides little positive curvature, so indefiniteness is likely to be broader than in this benchmark. Whether the ranking of the methods carries over remains to be tested. Finally, the QPs were solved in isolation with a Python prototype. An implementation in HPIPM and `acados` is needed for a run-time comparison, and, since the full step from the indefinite QP solution might be too aggressive, it remains to be investigated how the globalization mechanisms, such as a trust region and line search, should be designed for SQP with the indefinite QP solver for good closed-loop performance.

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